

TEMPLATE

KEY PROJECT INFORMATION & PROJECT DESIGN DOCUMENT (PDD)

PUBLICATION DATE **14.10.2020**

VERSION **v. 1.2**

RELATED SUPPORT

– **TEMPLATE GUIDE Key Project Information & Project Design Document v.1.2**

This document contains the following Sections

Key Project Information

Section A – Description of project

Section B – Application of approved Gold Standard Methodology (ies) and/or demonstration of SDG Contributions

Section C – Duration and crediting period

Section D – Summary of Safeguarding Principles and Gender Sensitive Assessment

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Q – Summary of Approved Design Changes (project specific)

KEY PROJECT INFORMATION

GS ID of Project	GS11173
Title of Project	Decentralized clean cooking for smallholder farmers in Mexico
Time of First Submission Date	21/05/2021
Date of Design Certification	TBD
Version number of the PDD	3.2
Completion date of version	13/04/2022
Project Developer	Buen Manejo del Campo S.A de C.V (Sistema.bio)
Project Representative	Buen Manejo del Campo S.A de C.V (Sistema.bio) Alexander B. Eaton alex@sistema.bio
Project Participants and any communities involved	Buen Manejo del Campo S.A de C.V (Sistema.bio) is both the participant and representative.
Host Country (ies)	Mexico
Activity Requirements applied	☒ Community Services Activities ☐ Renewable Energy Activities ☐ Land Use and Forestry Activities/Risks & Capacities ☐ N/A
Scale of the project activity	☐ Micro scale ☒ Small Scale ☐ Large Scale
Other Requirements applied	NA (No specific requirements for small scale projects)
Methodology (ies) applied and version number	Technologies and Practices to Displace Decentralized Thermal Energy Consumption, Version 3.1
Product Requirements applied	☒ GHG Emissions Reduction & Sequestration ☐ Renewable Energy Label ☐ N/A
Project Cycle:	☐ Regular ☒ Retroactive

Table 1 – Estimated Sustainable Development Contributions

Sustainable Development Goals Targeted	SDG Impact (defined in Error! Reference source not found.)	Estimated Annual Average	Units or Products
13 Climate Action (mandatory)	Emissions Reductions	59,236	VERs
3 Good health and wellbeing ¹	Beneficiaries reporting air inside their homes is cleaner AND Beneficiaries reporting better hygiene conditions	90%, 90%	%
5 Gender equality ²	- Women with access to technology - Reported time saved due to use of biogas - Share of women participating in the project	- 2,800 - 8.4 - 50	- Number of women - Hours per month per woman - %
7 Affordable and clean energy	Number of connections of clean and renewable energy source AND Number of persons getting access to a clean and renewable energy source	3,000 14,400	Number of connections Number of persons
8 Decent work and economic growth	Jobs generated Training offered to staff	20 40	Jobs Number of staff trained

¹ Initially, the PD considered the ADALYs product certification, however, due to the high cost of monitoring and the lack of financing for such certification, said product could not be incorporated. In case financing is secured, the project will seek ADALYs certification following the respective requirements, including the design change.

² Initially, the PD considered the Impact Gender Statement certification, as the guidance for this product and financing is secured, the project will seek Impact Gender Statement certification following the respective requirements, including the design change.

	Number of hectares fertilized with bio-fertilizer	7,537	Hectares
15 Life on land	Final use of bio-fertilizer	Sold; used by the beneficiary; or given free of charge	Final use

SECTION A. DESCRIPTION OF PROJECT

A.1. Purpose and general description of project

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Agriculture is responsible for a quarter of all greenhouse gas emissions, while climate change threatens global food production. Smallholder farmers are the most vulnerable in face of these challenges and often do not have access to technology, training or financing services to help them overcome them. The generation and use of biogas and biofertilizer leads to the reduction of greenhouse gas emissions by displacing conventionally used non-renewable fossil fuels, such as firewood or LP gas, by clean and renewable biogas, and by adding a management infrastructure of residues to properly process the manure generated in the farms.

Buen Manejo del Campo S.A de C.V (Sistema.bio) is a social enterprise that provides access to innovative biodigester technology, training and financing to address the challenges of poverty, food security, and climate change. Sistema.bio manufactures and distributes high-quality, affordable biodigesters that enable farmers around the world to convert waste into energy and fertilizer. By providing flexible interest-free repayment plans and comprehensive monitoring services, Sistema.bio works with farmers to ensure they become more sustainable, independent, and productive while mitigating the carbon emissions generated by agricultural activities. This project will provide a wide range of social, economic and environmental benefits for families and communities in Mexico through the installation of biodigesters.

The project includes 9 sizes of biodigesters to adjust to the different types of farmers. The project will focus on small and medium-scale producers in Mexico, specifically subsistence farmers who typically use woodfuel or LPG for their energy needs both in the household and productive uses in the farm, and who currently do not manage the manure generated by their animals and have either solid or liquid storage practices for this. The project will install 3,000 biodigesters over a period of 30 months in Mexico, with a focus on the Bajío, and the central and southeastern states of the country.

The technology used for this project is the Sistema.bio anaerobic digester package that provides a full digester, clean cooking and fertilizer management system. Sistema.bio is a prefabricated hybrid reactor that can be installed easily and becomes

a permanent piece of infrastructure for farmers to process the daily waste of a farm and livestock. This waste manure breaks down anaerobically, producing methane-rich biogas that can be used for providing clean cooking and thermal processes for the farm and household. Biogas flows to different points of use like cooking stoves, burners, etc. with a simple, long lasting gas line. The output of the digester is a powerful biofertilizer which is stored and applied in the fields as a substitute to chemical fertilizers. The pictures below show an installed system, and our two-burner spider stove.



Photo 1 – Sistema 16 installed in rural Mexico



Photo 2 - Sistema.bio Spider stove

The project comes with financing, guarantee and a full-service package of installation, training and after sales-service from local technicians. Installers are responsible for installing the unit, training the user and monitoring the system on scheduled visits to ensure long-term adoption. The program has a payment plan that works for farmers, from 6 to 24-month repayment periods, considering their cash flow and savings generated by the product. A loan appraisal survey is made via the digital Salesforce platform and evaluated by a credit committee. Collections are leveraged by field team activities.

However, there are many small farmers in the communities where this program is operating whose cash flow limits their ability to acquire a system even with the carbon funding support and Sistema.bio's flexible payment plans. Sistema.bio will work with local and national NGOs, cooperatives and associations to gain access to smallholder farmers and for the implementation of the project. Organizations participate in order to impact the participants in their initiatives and may be contracted for some services or may contribute resources to the project. Partnership agreements clearly define the carbon and impact benefits as remaining with Sistema.bio.

A.1.1. Eligibility of the project under Gold Standard

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The project falls under GG4GG Community Services Activity Requirements:

Eligible Project Types & Scope: The project leads to climate change mitigation by providing access to resources (biogas) to households and commercial institutions.

Types of project: The project falls under:

- 'Renewable energy'. The project is generating thermal energy with the biogas (waste to energy) for targeted users and/or applications.
- Waste management and handling: Management of animal waste (cattle and swine dung) to deliver biogas for targeted users and/or applications.

Project Location: Mexico

Project Area, Project Boundary and Scale: The project area and boundary include all Mexican states. The project boundary is identified as the physical, geographical sites of the project technologies are installed. (Project Area and Boundary is described under section A.2 below.)

This boundary also hosts the baseline and project fuel collection area.

The biodigesters and biogas stoves of the project will be uniquely identified in order to avoid double counting with overlapped activities in the project area.

No specific requirement due to the scale of the project.

Scale: For thermal applications of biomass, biofuels or biogas (e.g. cookstoves), the limit of 45 MW(th) is the installed/rated capacity of the thermal application equipment or device(s) (e.g. biogas stoves). ³ The rated capacity of the different appliances used in the project are listed below along with the calculated power capacity:

Appliance	Rated Capacity pe runit (kW)	Number of units	Total Power (MW)
Biogas stove	3.20	3,000	9.60
Campana	2.00	186	0.37
Water heater	8.70	186	1.62

³ Standard CDM project standard for project activities Version 03.0, Section '7.12.2.Project type and eligibility', § 119 (iii).

BioBurner 1 (flare)	3.40	40	0.14
BioBurner 20 (flare)	20.00	20	0.40
BioBurner 30 (flare)	30.00	20	0.60
BioBurner 40 (flare)	40.00	20	0.80
			13.53

The total power capacity of the project 13.53MW (th), is below the small-scale threshold.

Host Country Requirement: Sistema.bio's biogas programs are in complete compliance with Mexico's legal, environmental, ecological and social regulations. Specifically, in Mexico, there are no legal requirements for small-scale prefabricated biodigesters such as the ones used for this project. In 2010, the environmental federal agency, SEMARNAT, jointly with the agriculture federal agency, SAGARPA, published a document titled "Especificaciones Técnicas para el Diseño y Construcción de Biodigestores en México"⁴ that outlines the technical specifications for lagoon cover biodigesters, which do not apply to this project's biodigesters. All the norms referred in the document quoted are not *per-se* applicable to biodigesters but relevant for their components, stages of the waste generation, or stages of treatment process (e.g. solids/sludge waste handling, storage of flammable compounds). In addition, the technology offers a comprehensive waste management system that allows farms to comply with existing regulatory standards addressing limits to animal waste discharges in farms, such as NOM-001-SEMARNAT-1996, which establishes the maximum permissible limit for pollutants in residual waters discharged into national water sources. Sistema.bio allows farmers to capture waste streams into the system, where they are treated prior to being discharged as biofertilizer, which is captured in a tank at the end of the digester. Therefore, the system limits the residuals waters that are discharged into water sources.

Legal Ownership: The carbon rights transfer from project beneficiaries are collected transparently with full, prior, and informed consent (FPIC). The carbon transfer form will be made available for the design certification and at the performance review (see sample text in section A.1.2 below).

Official Development Assistance (ODA) Declaration: ODA Declaration submitted. The project is not registered with any other voluntary or compliance schemes.

The project is located in Mexico, which does not have an emission reduction cap enforced for the livestock and domestic sector nor the possibility to require obligation to reduce emissions under compliance schemes for said sectors. In addition, this

⁴ <https://www.biopasos.com/biblioteca/Manual%20captura%20metano%20agricultura%20ganaderia.pdf>

project does not overlap with the activities of another Gold Standard or other voluntary compliance standard programs.

The table below summarizes how the project meets the eligibility principles and requirements.

Eligibility Criteria, Section 4 -ELIGIBILITY PRINCIPLES AND REQUIREMENTS, Gold Standard PRINCIPLES & REQUIREMENTS, Version 1.2 – Effective 24th October 2019

Principle 1: Contribution to Climate Security & Sustainable Development

(a) ELIGIBLE PROJECT TYPES	The Project type is automatically eligible for Gold Standard Certification because there are GS approved Activity Requirements and Impact Quantification Methodologies associated with the project activity, which is community-based activity that includes End-use energy efficiency biogas stoves and Waste management and handling biodigesters.
(b) BASELINE AND PROJECT SCENARIOS	The further information about Baseline and Project scenarios can be found in sections B.4 and A.1 respectively of the PDD.
(c) CONTRIBUTION TO CLIMATE SECURITY & SUSTAINABLE DEVELOPMENT	The project's impact to Climate security and sustainable development is summarized below:
	SDG 3 – Reported improved cleaner air inside participants' homes and better hygiene conditions SDG 5 – Women with access to technology, Reported time saved due to use of biogas, and Share of women participating in the project SDG 7 - Number of connections, and number of persons getting access to a clean and renewable energy source. SDG 13 – Climate Action: 296,181 tCO₂e mitigated cumulatively during the entire crediting period. SDG 8 - Decent work and economic growth: 20 jobs generated, 40 staff trained SDG 15 – Life on land: 7,537 hectares fertilized with bio-fertilizer, and final use of bio-fertilizer (sold, used by the beneficiary, or given free of charge) Further information on Monitoring of SDG impact can be found in section B.6.3 below.

Principle 2: Safeguarding Principles

(a) SAFEGUARDING PRINCIPLES ASSESSMENT	The safeguarding principles are discussed in the 'Appendix 1 - Safeguarding Principles Assessment' below.
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Principle 3: Stakeholder Inclusivity

(a) STAKEHOLDER CONSULTATION & ENGAGEMENT	The Stakeholder Consultation is conducted in accordance with the standard requirements comprising one mandatory physical
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meeting, one virtual meeting⁵ and one stakeholder feedback round lasting for more than two months. These consultations are open to anyone wishing to attend. The project also established ongoing engagement process to ensure anyone interested in providing feedback or complains finds a communication channel to do so.

Principle 4: Demonstration of real outcomes:

(a) PROJECT START DATE	The project start date is set up as April 1 st 2021
(b) DEVELOPMENT OF MONITORING AND REPORTING PLAN	Please see section B.7.1 below

Principle 5: Financial Additionality & Ongoing Financial Need

(a) FINANCIAL ADDITIONALITY	The project activity is eligible under the Annex B - Positive list, 'GG4GG Community Services Activity Requirements Version 1.2', Annex B – Positive List for community service projects, paragraph 1.1.3.
(b) ONGOING FINANCIAL NEED	Ongoing financial need will be demonstrated at Design Certification Renewal.
(c) STACKING AND FINANCIAL ADDITIONALITY	NA. The project is not seeking to issue multiple different GS certified impact statements and products.

A.1.2. Legal ownership of products generated by the project and legal rights to alter use of resources required to service the project

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Implementation of the proposed project doesn't involve any activity that causes alteration of any resource; therefore acquiring any specific legal right to do so is not applicable. However, the entitlement of the emission reductions generated by the project shall be transferred to the project developer from the beneficiary households through a signed contract.

The translated text of the carbon rights waiver is shown below:

The Buyer, accepts the following regarding the use of the biodigester and stove "the good":

1. That they have the right to install "The good" in the designated site.

⁵ The virtual meeting took place on May 7th, 2021, in person meetings took place on November 17th in San Juan Tuxco, Puebla and November 19th, 2021 in Oxxutzcab, Yucatán.

- 2. That once installed, they agree to use "The good" in accordance with all the instructions and manuals provided by the installer technician.*
- 3. That they will use or burn all the biogas produced by the system.*
- 4. That they will add the indicated amount of fresh animal manure to the system.*
- 5. That the resulting biofertilizer will be used or provided for others to use.*
- 6. That it will commit to keep the system in use throughout its life.*

As part of this agreement, the Buyer transfers all its rights to carbon credits, or any other impact (including health, social, economic or environmental) resulting from the use of its Sistema.bio unit to the Seller and will not attempt to transfer or sell credits of carbon or any other impact to third parties. The Buyer agrees to cooperate with Seller in any effort to collect information or verify any data related to verification and monitoring related to carbon credits and impacts.

Samples of carbon rights waivers will be made available to the validator.

A.2. Location of project

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The project activity will be implemented in all parts of Mexico where the conditions to sell and install biodigesters are met. The project activity is mostly implemented in Hidalgo, Puebla, Estado de Mexico, Jalisco, and Yucatán where we have field offices.

The GPS coordinate and a unique identification will be recorded for all biodigester and biogas stoves installed. Sistema.bio method for tracking data is through a data tracking system which records data from smartphones in real-time in the field and tracks and analyzes this data throughout the client life cycle in a cloud-based dynamic database. Using the Taro Works smartphone platform, technicians collect data, GPS coordinates and photos directly in the field. This data is loaded onto the Salesforce platform.

The details of the current geographical location of the project are presented below:

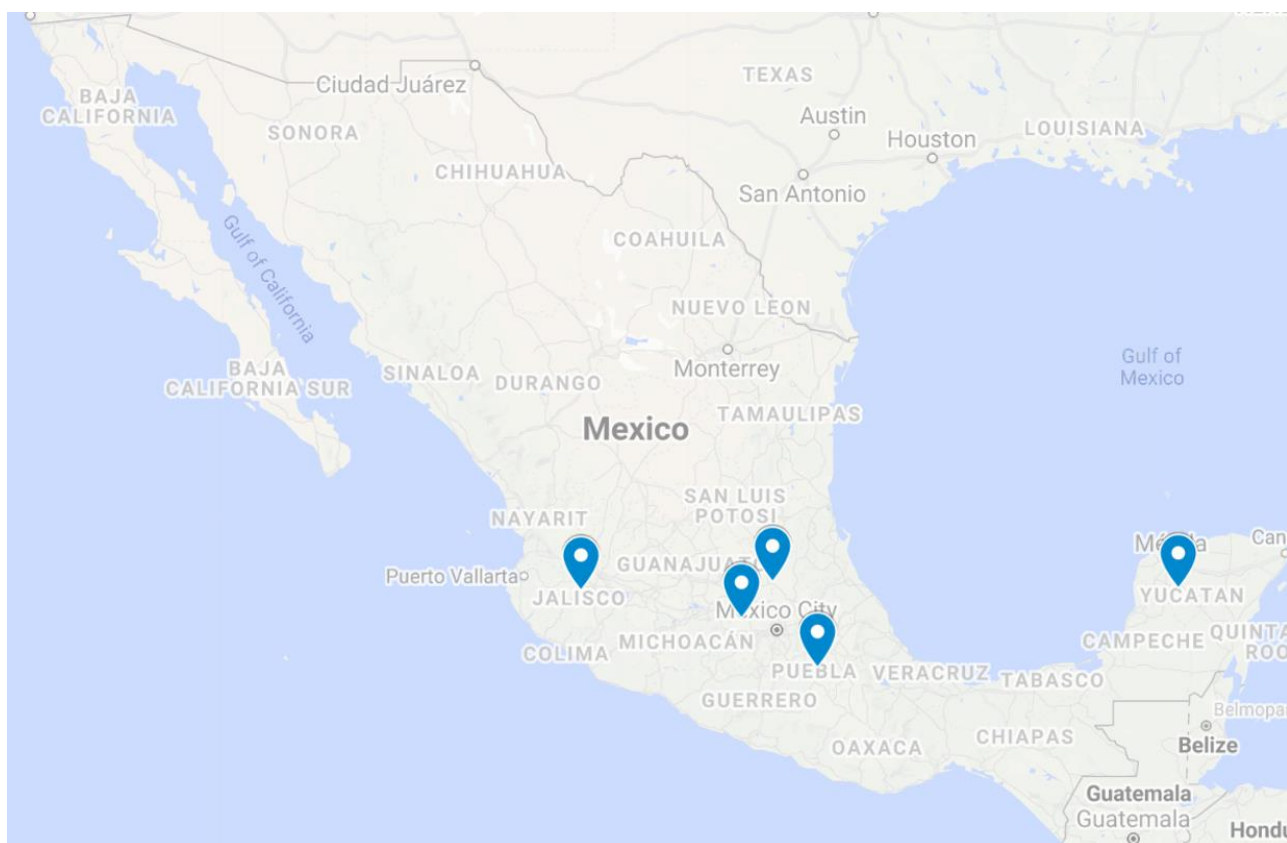


Figure 1 - Project locations

The project location is defined by the GPS coordinates of every biodigester and stove installed. Since it is not practical to include in the PDD all GPS points and most of them are not yet known because the installation is progressive and expected to take 30 months, the description of the states where the project expects to install units is considered adequate. The specific location of every biodigester and stove will be made available at the verification.

A.3. Technologies and/or measures

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This project is applicable for Gold Standard VER registry, and under Gold Standard methodology it is considered a biogas project.

The technology used for this project is the Sistema.bio anaerobic digester package that provides a full digester, clean cooking and fertilizer management system.

Sistema.bio is a prefabricated hybrid reactor that can be installed easily and becomes a permanent piece of infrastructure for farmers to process the daily waste of a farm and livestock. This waste manure breaks down anaerobically, producing methane-rich biogas that can be used for providing clean cooking and thermal processes for the farm and household. Biogas flows to different points of use like cooking stoves, burners, etc. with a simple, long lasting gas line. The output of the digester is a powerful biofertilizer which is stored and applied in the fields as a substitute to chemical fertilizers. The various components of the Sistema.bio reactor can be seen in the Figure 2 below:

COMPONENTS OF Sistema.bio®

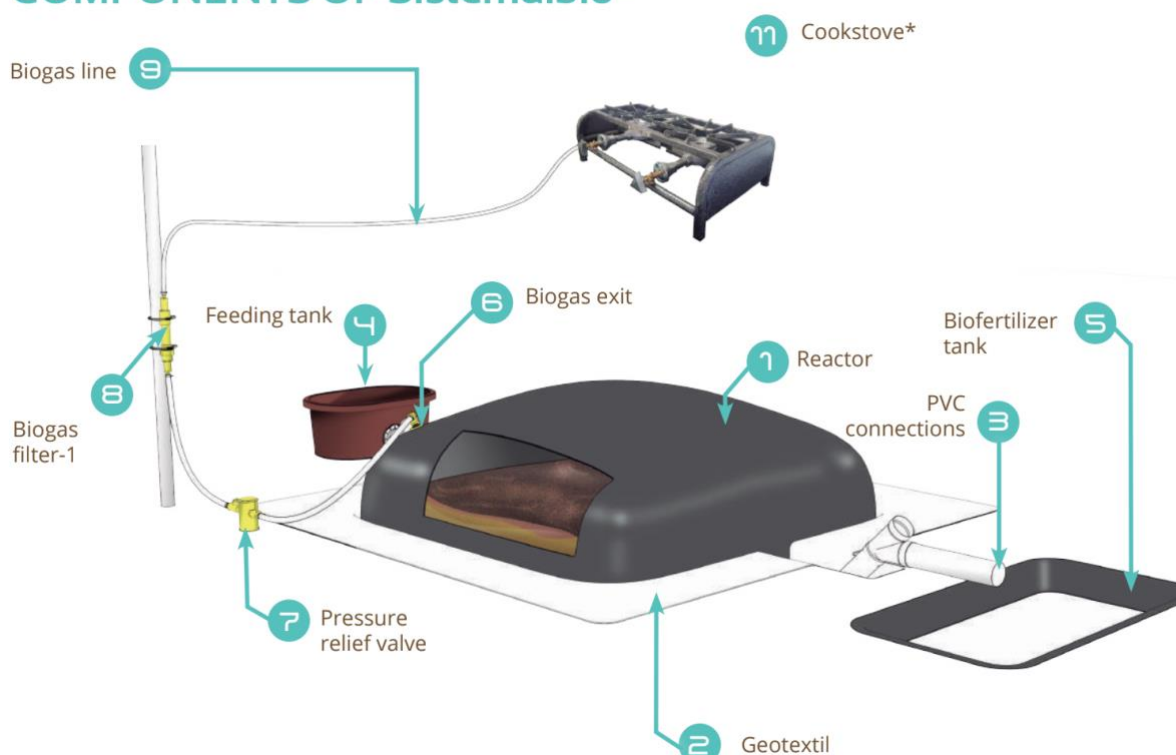


Figure 2 Sistema.bio offers a proven and patented complete biogas kit that includes a high quality hybrid reactor and all the connections and accessories necessary for farmers to begin cooking on biogas within weeks.

The Sistema.bio Stove has been tested in accordance with ISO 19867-1 T, with a thermal efficiency of 40.3%/50% and 4.5 kW/3.4 kW of cooking power depending on the model used (see below the model specifications). In addition, the products are designed to have a long useful lifetime in the harsh, outdoor environments of smallholder farms throughout the world, with a focus in tropical or semi-tropical climates. The reactors are made from high density polyethylene blends that will retain mechanical and chemical properties in fully exposed conditions for an estimated 20-40+ years. This data has been validated through field data, laboratory exposure testing, certified material characteristics and supplier claims. The rest of the

Sistema.bio components of the kit can be replaced or repaired locally at low costs. Materials that may not last 20 years, like hose clamps, PVC piping or gas tubing can be easily replaced on the unit from local materials with local know-how and with technical support as needed. In addition, most of the components, including the reactor, can be repaired or replaced in the field, ensuring continued operations over time.



BioStove-1

30.08.2020

Product description

The BioStove-1 is a thermal domestic appliance. It can be used daily to cook at home.

Basic specs

Production place	Ferro - Mexico
Weight	4.3 kg
Size (h x l x w)	11 x 52 x 30 cm
Materials	Carbon steel, electrostatic painting, aluminium



Additional specs

Biogas connection type	1"
Burner material	aluminium
Ignition mode	Manual
Minimum work pressure	0.5mbar
Estimated life time (years)	3 with BioGasPurifier-1
Minimum system size	Sistema 6

Consumption and performances

Average biogas consumption	0.6m ³ /h with two burners ON
Maximum time use, with minimum sistem size	2 hours a day with a Sistema 6 and two burners ON
Performances (description)	Each burner has an average 1.7 kW firepower The stove has an average total firepower of 3.4 kW
Maximal efficiency	50%, similar range than LP stoves (WBT)

Figure 3 - Specifications Biogas Stove Model 1

Product description

The BioStove-2 is a thermal domestic appliance. It can be used daily to cook at home.

Basic specs

Production place	Ferro - Mexico
Weight	6kg
Size (h x l x w)	19 x 58 x 36 cm
Materials	Carbon steel, electrostatic painting, aluminium

Additional specs

Biogas connection type	1"
Burner material	aluminium
Ignition mode	Manual
Minimum work pressure	0.5mbar
Estimated life time (years)	3 with BioGasBurner-1
Minimum system size	Sistema 6



Consumption and performances

Average biogas consumption	0.8m ³ /h with two burners burning gas
Maximum time use, with minimum sistem size	1.5 hour a day with a Sistema 6 and two burners ON
Performances (description)	Each burner has an average 2.25 kW firepower The stove has an average total firepower of 4.5 kW
Maximal efficiency	43%, similar range than LP stoves (WBT)

Figure 4 - Specifications Biogas stove model 2⁶

Measured against the baseline conditions of the farm and household, clean energy displaces existing energy costs and time collecting firewood^{7 8 9 10}, and also reduces health risks for women in the household linked to poor indoor air quality^{11 12}. Fertilizer improves access to nutrients for plants and soil base, reduces costs and health and

⁶ The estimated lifetime is based on a conservative estimate given that this stove model has been in use for a short period of time; field results of the stoves installed to date suggest that they can last more than five years.

⁷ Mwirigi, J. P. (2009). Socio-economic Constraints to Adoption and Sustainability of Biogas Technology by Farmers in Nakuru Districts, Kenya. *13*(2), 106-115.

⁸ Rupf, G. P. (2016). Broadening the Potential of Biogas in Sub-Saharan Africa: An Assessment of Feasible Technologies and Feedstocks. *Renewable and Sustainable Energy Reviews*, *61*, 556-571.

⁹ Kabir, H. R. (2013). Factors Determinant of Biogas Adoption in Bangladesh. *28*, 881-889.

¹⁰ Mengistu, M. B. (2016). Factors Affecting Households' Decisions in Biogas Technology Adoption, the Case of Ofla and Mecha Districts, Northern Ethiopia. *93*, 215-227.

¹¹ Mengistu, M. B. (2015). A Review on Biogas Technology and Its Contributions to Sustainable Rural Livelihood in Ethiopia. *68*, 306-316.

¹² Ghimire, P. (2013). SNV Supported Domestic Biogas Programs in Asia and Africa. *49*, 90-94.

environmental damage from chemical fertilizers^{13 14 15}. By adding waste treatment to the household, odors, flies, and water contamination risks are greatly reduced¹⁶. Using the SDG framework to link these outputs to further impacts, our technology impacts: poverty (through increased incomes and decreased costs); hunger (improved food production through organic fertilizer); health (reduced risks from smoke and waste with biogas); energy access (clean, sustainable long-term fuel source), and; climate change (reduced emissions from waste, energy and fertilizers).

COMPARATIVE DIAGRAMS

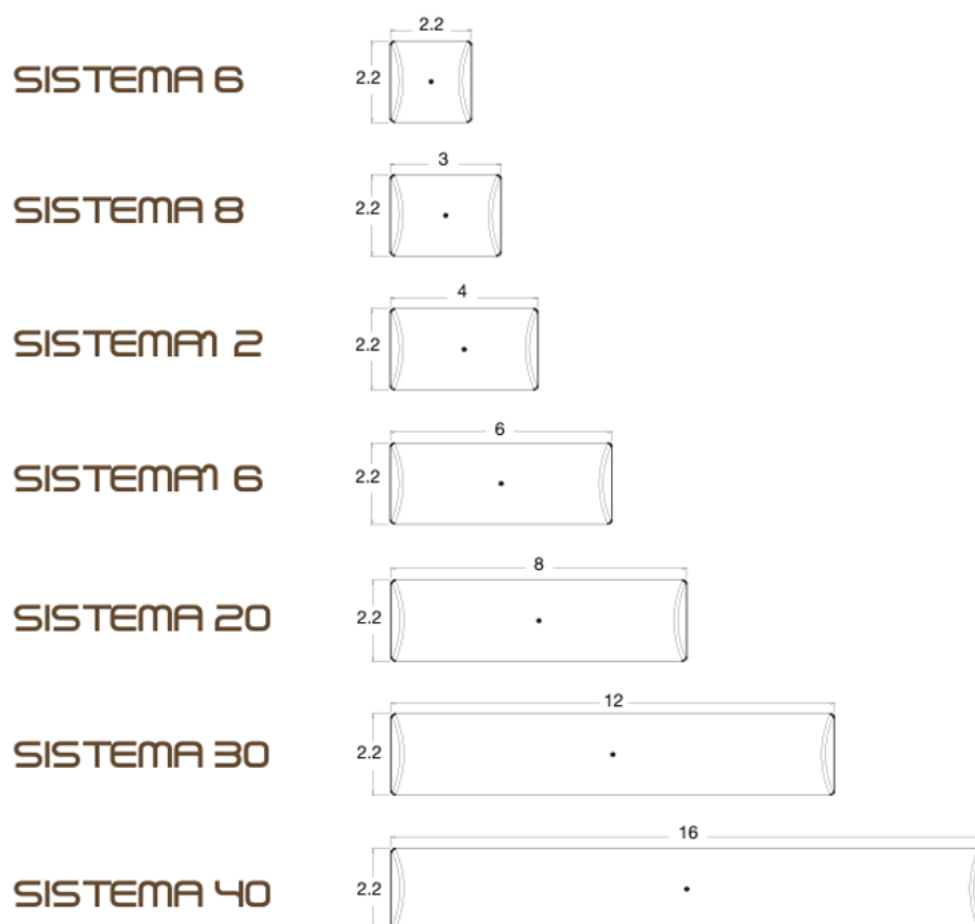


Figure 5 - Available sizes for the Sistema (measures provided in meters)

¹³ SNV. (2015). *Bioslurry Always Means Profit. Success Stories from the Tanzania Domestic Biogas Program*. Retrieved from https://snv.org/cms/sites/default/files/explore/download/bioslurry_always_means_profit_-_success_stories_from_the_tanzania_domestic_biogas_program1.compressed.pdf

¹⁴ PNB-BF. (2016). *Analyse Comparee des Teneurs en Elements Fertilisants entre la Bouse et L'effluent et Leurs Performances Agronomique et Economique*.

¹⁵ Hivos. (2014). Retrieved from Bioslurry: A Supreme Fertilizer: <https://www.ourenergypolicy.org/wp-content/uploads/2014/05/bioslurry.pdf>

¹⁶ Ibid.

The table below provide further details of the capacity for the different sizes of biodigesters to be implemented.

Type	Capacity (m ³)
Sistema 6	6
Sistema 8	8
Sistema 12	12
Sistema 16	16
Sistema 20	20
Sistema 30	30
Sistema 40	40
Sistema 80 (2 x Sistema 40)	80
Sistema 120 (3 x Sistema 40)	120

Reactors' capacity

A.4. Scale of the project

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For thermal applications of biomass, biofuels or biogas (e.g. cookstoves), the limit of 45 MW(th) is the installed/rated capacity of the thermal application equipment or device(s) (e.g. biogas stoves).¹⁷ The rated capacity of the biogas stove is 3.2 kW.¹⁸ The Project includes installation of 3,000 units which result in a total capacity of 9.6MW(th), far below the small-scale threshold.

A.5. Funding sources of project

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The project does not receive public funding. The project is implemented by the client. Carbon waiver will be signed by the project owner and carbon rights will be available with Buen Manejo del Campo S.A de C.V (local entity).

¹⁷ Standard CDM project standard for project activities Version 03.0, Section '7.12.2.Project type and eligibility', § 119 (iii).

¹⁸ Emission and Performance Report based on ISO 19867. See file 'Sistema.bio Bioga_2 ISO Cert blue stove.pdf'

SECTION B. APPLICATION OF APPROVED GOLD STANDARD METHODOLOGY (IES) AND/OR DEMONSTRATION OF SDG CONTRIBUTIONS

B.1. Reference of approved methodology (ies)

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The relevant project type and category is: Technologies and Practices to Displace Decentralized Thermal Energy Consumption Version 3.1 Reference:

https://globalgoals.goldstandard.org/standards/407_V3.1_EE_ICs_Technologies-and-Practices-to-Displace-Decentralized-Thermal-Energy-TPDDTECConsumption-.pdf

Other methodological tools applied:

Calculation of the fraction of non-renewable biomass Version 03.0

(<https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-30-v3.0.pdf>)

B.2. Applicability of methodology (ies)

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	Criterion	Applicability
1.	The project boundary needs to be clearly identified , and the technologies counted in the project are not included in any other voluntary market or CDM project activity (i.e. no double counting takes place). In some cases there maybe another similar activity within the same target area. Project proponents must therefore have a survey mechanism in place together with appropriate mitigation measures so as to prevent any possibility of double counting.	The project boundary is clearly identified, the project is not included in any other voluntary or compliance markets and Sistema.bio tracks unique GPS and contact information of every biodigester that is installed through a digital management platform to prevent double counting. ¹⁹
2.	The technologies each have continuous useful energy outputs of	The maximum energy output of the biodigesters is in range of 7 to 48 kW _{th}

¹⁹ There is only another active project identified under CAR the same project area. This project is located in Yucatan stated and is developed in a farm belonging to group Kent and Sorensen Overseas, SA de CV, which is clearly identified as a distinct project (see <https://thereserve2.apx.com/mymodule/reg/TabDocuments.asp?r=111&ad=Prpt&act=update&type=PRO&aProj=pub&tablename=doc&id1=815>)

Under CDM there are a couple of projects and one CPA in similar zones with similar activity. Those projects are developed by different groups e.g. Agrícola Europa, S.P.R. de R.L; Rancho Covadonga S.A. de C.V.; Kent and Sorensen Overseas, SA. de CV, clearly different from Sistemabio. (see [https://cdm.unfccc.int/filestorage/T/V/W/TVWL1NOA3RCEY9KJ2GDZ8MSP6HI4B5/PDD.pdf?t=Tml8cjFsaHNtFDBqui9AptXQG_MGEGee8f2B](https://cdm.unfccc.int/filestorage/T/V/W/TVWL1NOA3RCEY9KJ2GDZ8MSP6HI4B5/PDD.pdf?t=Tml8cjFsaHNtFDBqui9AptXQG_MGEGee8f2B;); https://cdm.unfccc.int/filestorage/g/p/LRUAYM9C3E05HPX7TOZ8DBI4Q1JFN6.pdf/PDD_AWMS%20Methane%20Recovery%20Project%20in%20Tlaquiltenango%20Morelos_re.pdf?t=TEI8cjFsaHNwfdAz66e8zAdiX3htvBqwtMDI; <https://cdm.unfccc.int/filestorage/g/z/2KJBRLO6ZSU9CWHN58M7DIYX1GQPEF.pdf/CPA-DD-9337-SSC-10-09.pdf?t=RGR8cjFsaTBifDAV8dtGx-D1BK-mOSe1H-Fj>)

	less than 150kW per unit: (defined as the total useful energy delivered from start to end of operation of a unit divided by time of operation). For technologies or practices that do not deliver thermal energy in the project scenario but only displace thermal energy supplied in the baseline scenario, the 150kW threshold applies to the displaced baseline technology.	based on the system size,²⁰ which is below the 150kW per unit threshold. The Sistema.bio stove has been tested in accordance with ISO 19867-1 T, with a thermal efficiency in a range of 40.3% to 50%, a 90%+ methane destruction efficiency and 3.4 kW to 4.5 kW cooking power,²¹ which is less than 150kW per unit.
3.	The use of the baseline technology as a backup in parallel is permitted as long as a mechanism is put into place to encourage the removal of the old technology and the definitive discontinuity of its use.	Sistema.bio will monitor use of baseline technology regularly. Our team will conduct visits to users after installation to get feedback on the use of baseline and new technology and understand the reasons for the use of baseline technology, solve any doubts or hesitations.
4.	The project proponent must clearly communicate to all project participants the entity that is claiming ownership rights of and selling the emission reductions resulting from the project activity.	The end user of each biodigester will confirm that they transfer the ownership of VERs to the project developed. Sample Copy of agreement with the end users is provided in a separate document.
5.	Project activities making use of a new biomass feedstock in the project situation (e.g. shift from non-renewable to green charcoal, plant oil or renewable biomass briquettes) must comply with relevant Gold Standard specific requirements for biomass related project activities, as defined in the latest version of the Gold Standard rules.	The project activity does not involve usage of any new biomass feedstock. Thus, this condition is not applicable to the project.

²⁰ Based on the maximum flow rate capacity of the biogas pipe used for all the systems.

²¹ Based on the two biogas models available. See stoves specification in section A.3 above

The project also follows the applicability condition described in the Annex 6 of the TPDDTEC methodology which states. The following:

If more than one climate zone is included in the project activity, a distinction per climate zone must be considered.²² The distinct geographical boundary of each project area must be clearly documented in the project documentation, using representative GPS data.

The project has identified at least two climate zones: temperate and warm. Each area will be documented using GPS coordinates to corroborate its location and data from official sources about mean annual temperature.

In addition, the project follows the following conditions:

1. Adequate evidence to demonstrate that indoor air pollution (IAP) levels are not worsened compared to the baseline, and greenhouse gases (as listed in section 2.1) emitted by the project fuel/stove combination are estimated with adequate precision.
 - a. The project will improve IAP in households that switch from wood fuel to biogas produced by the biodigesters. Our monitoring procedures will track this in order to verify usage. Biogas is a clean energy that reduces PM2.5 levels²³, therefore our technology improves, rather than worsens indoor air pollution in households that adopt the biodigesters.
 - b. As per applied the GS TPDDTEC methodology version 3.1, project emissions will be accounted for the activities below, using the methodology described in section B.6.1:
 - i. CO2 Emissions due continued use of the old technology.
 - ii. Methane emission from the biodigester which includes physical leakage and incomplete combustion of the biogas, as well as emissions from the animal waste not treated in the biodigester.
2. Records of renewable fuel sales may not be used as sole parameters for emission reduction calculation but may be used as data informing the equations in section 2.0 of this methodology.
 - a. The project will not introduce sales of renewable fuel. No further requirements are needed.

B.3. Project boundary

²² Currently two climate zones have been defined: Temperate and Warm. The mean annual temperature from official sources has been used to identify climate zones. For examples: Yucatan > 25°C identified as warm (n/yuc/territorio/clima.aspx); Jalisco and Hidalgo 15 to 25 °C (<http://www.cuentame.inegi.org.mx/monografias/informacion/jal/territorio/clima.aspx?tema=me&e=14>; <http://cuentame.inegi.org.mx/monografias/informacion/hgo/territorio/clima.aspx?tema=me&e=13>)

²³ Buysman, E. (2015) Biogas and Household Air Quality: Study on Household Air Quality and Estimated Health Improvement of users of Biogas Stoves versus Wood-fired Stoves in Rural Cambodia. Hivos. Retrieved from: https://hivoscarboncredits.org/assets/2016/03/30OCT2015_Biogas_and_household_Air_Pollution_Cambodia_FINAL_complete.pdf

>>

The project boundary is the physical, geographical site of the use of the project technology (biodigester and biogas stove) throughout Mexico. It incorporates all the physical geographical sites of where Sistema.bio biodigesters are installed. A total of 3,000 biogas plants will be installed by Sistema.bio at various sites throughout the geographical boundary of Mexico. Same number of biogas stoves will be installed.

Figure 6 below shows the installation process of the technology at the project site:

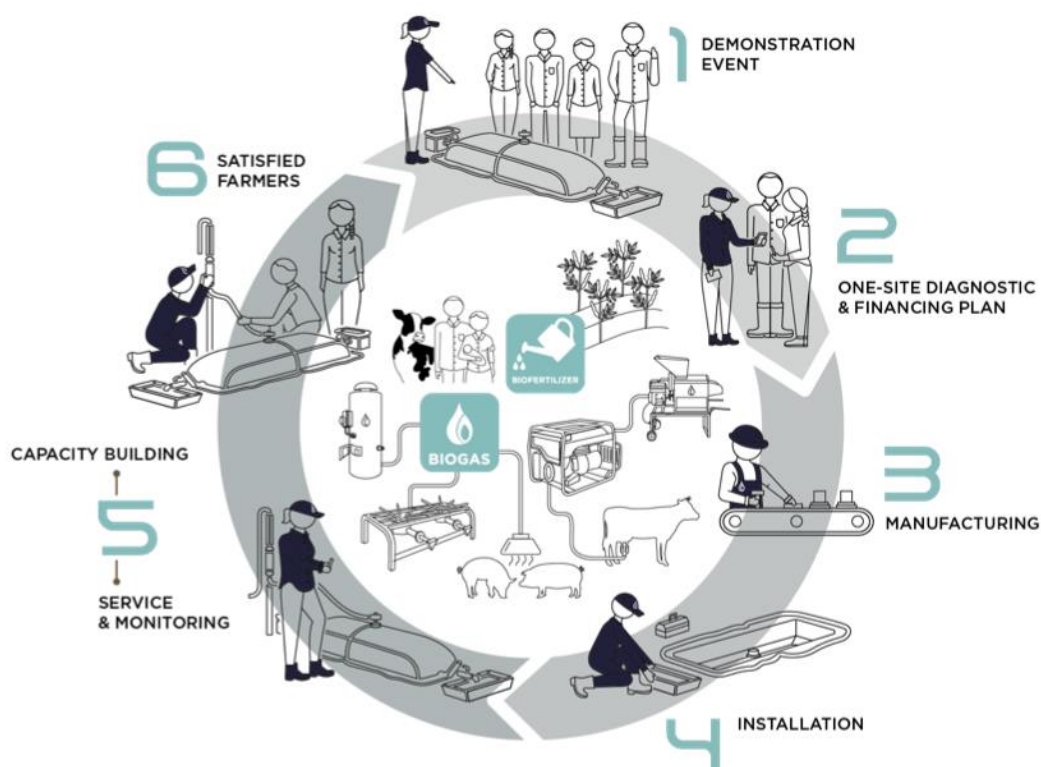


Figure 6 - Sistema.bio has used a proven process and methodology to track farmers through the promotion, acquisition and adoption process.

The emissions accounted from the various sources in the physical boundary of the project activity are as follows:

	Source	GHGs	Included ?	Justification/Explanation
Baseline	Thermal energy need	CO ₂	Yes	The major source of emissions in the baseline due to burning of firewood and use of LPG

Project scenario		CH ₄	Yes	Minimal source of emissions but included in the calculations.
		N ₂ O	Yes	Minimal source of emission but included in the calculations
	Animal waste handling and storage	CO ₂	No	Not Available, as baseline emissions from “feed” are not considered
		CH ₄	Yes	The major source of emission in baseline due to open dumping of animal manure
		N ₂ O	No	Not Available, as baseline emissions from “feed” are not considered
	Thermal energy need	CO ₂	Yes	Applicable for the project activity
		CH ₄	Yes	Applicable for the project activity
		N ₂ O	Yes	Applicable for the project activity
	Direct emissions from CO ₂ the biodigester	CO ₂	No	Excluded as CO ₂ emissions from biogas incineration are CO ₂ neutral
		CH ₄	Yes	Physical leakage of biodigesters is included as part of the project emission
		N ₂ O	No	Excluded for simplification

Leakages sources are described in section B.6.2

B.4. Establishment and description of baseline scenario

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In Mexico, woodfuel is the most prevalent energy source in rural areas, and it accounts for about 40% of the total residential energy consumption in the country, and Liquefied petroleum gas (LPG) is the other main fuel in the residential sector in

the country²⁴. The heat map in Figure 7 below shows where woodfuel is more prevalent.

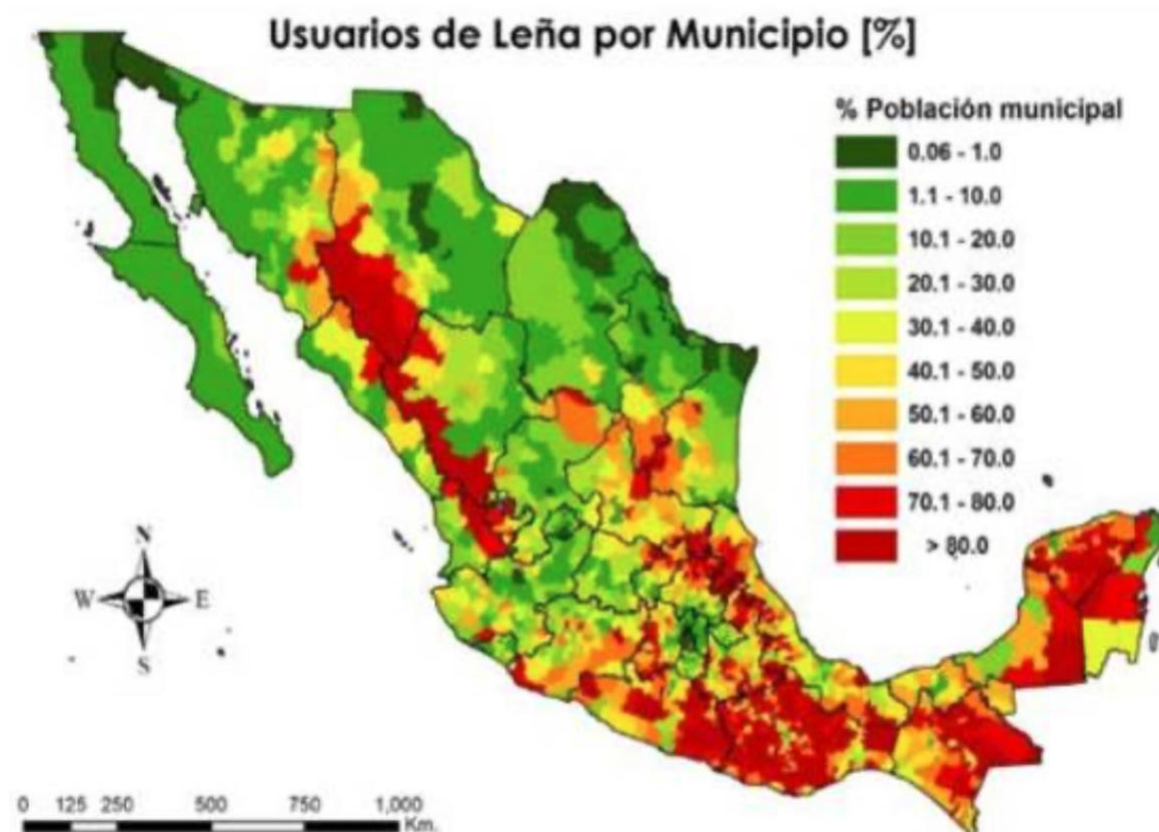


Figure 7 Heat map of woodfuel users in Mexico, 2017. Comisión Federal para la Protección contra Riesgos Sanitarios

The baseline scenario for small and medium holder dairy and swine farmers in Mexico is built from two main sources:

- a. The use of non-renewable fuels to meet energy requirements in the home and farm: The baseline energy consumption in rural Mexico is woodfuel and the fossil fuel LP Gas. Per capita consumption of fuel wood at rural households is from 1 – 3 kg/day/person. Typical energy needs can be met by about 40 kgs of LP Gas per month. We will use the TOOL 30: Calculation of the fraction of non-renewable biomass Version 03.0 to calculate the FNRB in Mexico.

As can be seen in the studies quoted below, the firewood has been the main source of energy for covering the energy need in for the rural population.

²⁴ Serrano-Medrano, M. G. (2019). Fuelwood use patterns in Rural Mexico: a critique to the conventional energy transition model. *Historia Agraria*, 77, 81-104.

"Approximately 27 million rural people in Mexico use biomass for cooking, where fuelwood represents approximately 80% of energy used by rural households and 50% of total energy use in rural communities."²⁵

"Currently, about one-fourth of Mexican households (27.2 million people) cook with fuelwood, either exclusively (18.7 million people) or in combination with LPG (8.5 million)."²⁶

"Serrano-Medrano et al. (2014) have projected a decrease by only 9 % of the number of exclusive fuelwood users and their associated fuelwood consumption between 2010 and 2030."²⁷

This scenario has not changed significantly in the last 15 years as it is stated in the National survey of the 'First National Survey On Energy Consumption In Particular Houses'²⁸ where it is stated textually (own translation):

"USE OF FUEL The largest proportion of the thermal energy consumed at households is devoted to cooking/heating food. The main fuel is LP gas with 79%. Followed by the use of firewood or charcoal with 11%, and natural gas with 7 percent."

In the same survey, it is motioned the following (own translation):

"It is noteworthy that although the use of firewood or charcoal for cooking still represent a significant proportion in the country, the use of improved cookstoves with

²⁵ Berrueta, Víctor M., et al. "Energy Performance of Wood-Burning Cookstoves in Michoacan, Mexico." *Renewable Energy*, vol. 33, no. 5, 2008, pp. 859–870., doi:10.1016/j.renene.2007.04.016.

²⁶ Masera, Omar R., et al. "From Cookstoves to Cooking Systems: the Integrated Program on Sustainable Household Energy Use in Mexico." *Energy for Sustainable Development*, vol. 9, no. 1, 2005, pp. 25–36., doi:10.1016/s0973-0826(08)60480-9.

²⁷ Berrueta, Víctor M., et al. "Promoting Sustainable Local Development of Rural Communities and Mitigating Climate Change: the Case of Mexico's Patsari Improved Cookstove Project." *Climatic Change*, vol. 140, no. 1, 12 Oct. 2015, pp. 63–77., doi:10.1007/s10584-015-1523-y.

²⁸ Primera Encuesta Nacional Sobre Consumo De Energéticos En Viviendas Particulares (Nov. 07th 2018) Comunicado De Prensa Núm. 541/18, INEGI (National Institute of Statistic and Geography for its meaning in Spanish) [Document available at(open on 14 Feb. 2020): <https://www.inegi.org.mx/contenidos/saladeprensa/boletines/2018/EstSociodemo/ENCEVI2018.pdf>

chimney for wood fuel or charcoal are practically non-existent. Which remains worrying given the conditions for cooking with these fuels and damage to health.”²⁹

According to the official latest statistics available from the Ministry of Energy³⁰ (SENER for its meaning in Spanish), fuelwood represents, on average in the latest three years, 33.10% of the total energy consumed at the residential sector. See table below:

Energy Information System						
Ministry of Energy						
National Energy Balance: Energy consumption in the residential, commercial and public sectors						
(petajoules)						
ACTUAL						
	2016	%	2017	%	2018	%
Residential Total Energy	756.85	1.00	751.61	1.00	760.60	1.00
Solar	6.10	0.01	6.42	0.01	7.11	0.01
Wood	251.56	0.33	250.31	0.33	249.08	0.33
Total petroleum (LPG, Kerosene)	249.47	0.33	246.45	0.33	246.45	0.32
Dry Gas	37.45	0.05	35.48	0.05	30.16	0.04
Electricity	212.28	0.28	212.95	0.28	227.80	0.30

It is clear that wood as a source of energy for covering cooking needs in the residential sector, especially in the rural context, is still very relevant. A transition to cleaner and more efficient technology is not expected in the short term.

- b. GHG emissions that result from poorly managed animal manure, which ranges from solid storage (MCF of 4%) to anaerobic lagoons (MCF of 78% at 20 °C). Farmers also use liquid storage (MCF 26-42%), deep bedding and pit storage (MCF 42% at 20 °C). All storage types include periods of longer than 30 days,

²⁹ Idem.

³⁰ Energy Information System, Ministry of Energy, National Energy Balance: Energy consumption in the residential, commercial and public sectors (petajoules). Available in the following link: (opened on 02 March 2020)
<http://sie.energia.gob.mx/bdiController.do?action=cuadro&cvequa=IIE4C03>

and generally manure is moved or pumped a couple times per year. This baseline information will be captured during the diagnostics of each farm prior to the installation of a biodigester and will follow the 2006 IPCC Guidelines for National Greenhouse Gas Inventories.

According to the latest GHG national inventory,³¹ the emission in the livestock sector corresponded to 72.47 MtCO₂e. These emissions represent the 9.88% of the total emission in the country. 67.92% corresponds to methane emission (CH₄), 4.55 mtCO₂e to Nitrous Oxide (N₂O). Out of the total emission, 75% are attributable to enteric fermentation and 25% to manure management, this last one, represent 19.75 mtCO₂.

The distribution of the emission from the manure management according to the national inventory is as follows:

Source	Emission (MtCO ₂)
Cattle	12.193
Swine	5.092
Other ³²	0.532

As can be seen, the emission from animal waste still represents a significant share of the emission in the country. The use of biodigesters has not been a common practice that help to lower the emission the livestock sector. Besides the biodigesters implemented as part of the CDM and other carbon standard projects, use of this technology has not been extended as a common practice and still represent a huge potential for mitigation.³³ In the past, there was a government initiative called FIRCO (*Fideicomiso de Riesgo Compartido* in Spanish) that provided financing for a limited number of large-scale biodigesters (73 units)³⁴. FIRCO trust is no longer in operation and given the modest number of biodigesters implemented in comparison with the

³¹ <https://www.gob.mx/inecc/acciones-y-programas/inventario-nacional-de-emisiones-de-gases-y-compuestos-de-efecto-invernadero>

³² Sheep, goat, horses, Mules and donkeys

³³ GIZ, Créditos de compensación para el Sistema de Comercio de Emisiones mexicano: Análisis de criterios para la priorización de proyectos (https://www.giz.de/en/downloads/giz2020_es_creditos_compensacion_sce_mx.pdf)

³⁴ <https://ecotec.unam.mx/wp-content/uploads/Diagnostico-Nacional-de-los-Sistemas-de-Biodigestion.pdf>

large demand in the country, it is concluded biodigester technology is not a common practice in the country.

In summary, the the baseline scenario is defined as (a) The continuation of the consumption of non-renewable biomass (firewood) and/or LPG for domestic purposes and (b) The release of CH₄ from the open storage and poorly manure management.

B.5. Demonstration of additionality

B.5.1. Prior Consideration

Specify the methodology, activity requirement or product requirement that establishes deemed additionality for the proposed project (including the version number and the specific paragraph, if applicable).

'GG4GG Community Services Activity Requirements Version 1.2', Annex B – Positive List for community service projects, paragraph 1.1.3

Describe how the proposed project meets the criteria for deemed additionality.	This project meets the third criteria because they project activity is solely composed of isolated units where the users of the technology/measure are households or communities or institutions and where each unit results in <= 600 MWh of energy savings per year or <=600 tons of emission reductions per year.
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As per GS4GG rule for retroactive projects, project documents need to submit to GS for the preliminary review within one year of the project start date to meet prior consideration. Particularly, the start date is April 1st, 2021, and are submitting the initial project documents to GS on May 21st 2021. Therefore, the project meets the prior consideration requirements.

B.5.2. Ongoing Financial Need

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The 'Principle:5 Financial Additionality & Ongoing Financial Need from the GS Principles and Requirements' v1.2, states textually "4.1.52 Ongoing Financial Need

shall be demonstrated at Design Certification Renewal". The project will meet said requirement at the renewal stage.

B.6. Sustainable Development Goals (SDG) outcomes

Relevant Target/Indicator for each of the three SDGs

Sustainable Development Goals Targeted	Most relevant SDG Target	SDG Impact
		Indicator (Proposed or SDG Indicator)
SDG 13 Climate Action (mandatory)	13.2, 13.3	13.2.1, 13.3.1 59,236 tCO ₂ e reduced annual average
SDG 3 Good health and wellbeing	Target 3.9 by 2030 substantially reduce the number of deaths and illnesses from hazardous chemicals and air, water, and soil pollution and contamination	Reported cleaner air inside homes and better hygiene conditions
SDG 5 Gender equality	Target 5.4 Recognize and value unpaid care and domestic work through the provision of public services, infrastructure and social protection policies and the promotion of shared responsibility within the household and the family as nationally appropriate	Women with access to technology, reported time saved due to use of biogas, share of women participating in the project
SDG 7 Affordable and clean energy	Target 7.1: By 2030, ensure universal access to affordable, reliable and modern energy services Target 7.2: By 2030, increase substantially the share of renewable energy in the global energy mix	Number of connections of clean and renewable energy source AND Number of persons getting access to a clean and renewable energy source
8 Decent work and economic growth	Target 8.5: By 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value	Number of jobs generated and trainings offered to staff
15 Life on land	Target 15.3: By 2030, combat desertification, restore degraded land and soil, including land affected by	Number of hectares fertilized with bio-fertilizer

desertification, drought and floods, and strive to achieve a land degradation-neutral world	Final use of bio-fertilizer
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B.6.1. Explanation of methodological choices/approaches for estimating the SDG Impact

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Methodological action to estimate SDG 13 outcomes:

Emissions Reductions

As per “Technologies and Practices to Displace Decentralized Thermal Energy Consumption Version 3.1” (GS TPDDTEC 3.1), the emission reductions considered for this project (methane emissions and woodfuel/LPG displacement) will be accounted from the following two sources:

(i) Methane emissions from Manure Management:

Baseline emissions due to Manure management have been accounted using IPCC TIER 2 approach as described in Annex 6 of the GS TPDDTEC 3.1 methodology. The following equation has been used:

$$BE_{awms,h} = GWP_{CH4} * \sum_T (EF_{awms(T)} * N_{(T),h})$$

Equation 1

Where:

$BE_{awms,h}$ = The baseline emission from handling of animal waste in premise h (tCO₂e per year)

GWP_{CH4} = 28 (value used for emission reduction achieved from 01/01/2021 onward. Aligned with IPCC AR5.)³⁵

$N_{(T)h}$ = The number of animals of livestock species per category T

$EF_{awms,T}$ = Emission factor for the defined livestock population category T, (tonCH₄ per head per year)

³⁵ Based on the GS Rule Update ‘Applicability of Global Warming Potential For Gold Standard For The Global Goals Projects’, Publication Date 03/06/2021.

Emissions factor calculation

- IPCC Tier 1: This approach is applicable to situations, where baseline data required for an estimation of the methane emission factor per category of livestock are NOT available.
- IPCC Tier 2: This approach is applicable to situations, where baseline data for an estimation of the methane emission factor per category of livestock are available. Examples for such a situation are when animals are kept in a confined area and the manure is collected following a specifically designed system.

The emissions factor $EF_{awms,T}$ for tier 2 approach is calculated using the following equation:

$$EF_{awms(T)} = VS_{(T)} * 365 * \left[B_{O(T)} * D_{CH_4} * \sum_k \frac{MCF_{BL,k}}{100} * MS_{(T,k)} \right]$$

Equation 2

Where:

$EF_{awms,T}$	= CH4 emission factor for livestock category T (tCH4 per animal per year)
$VS_{(T)}$	= Daily volatile solid excreted for livestock category T, (kg dry matter per animal per day)
365	= Basis for calculating annual VS production, (days per year)
$B_{O(T)}$	= Maximum methane production capacity for manure produced by livestock category T, (m ³ CH4 per kg of VS excreted)
D_{CH_4}	= Methane Density (0.00067 t per m ³ at room temperature (20 °C) and 1 atm pressure)
$MCF_{(BL,k)}$	=Methane conversion factors for the animal waste handling system in the baseline situation, by climate zone k, (%)
$MS_{(T,S,k)}$	=Fraction of livestock category T's manure treated in animal waste management system, in climate region k (dimensionless)

(ii) Carbon dioxide emissions from the combustion of non-renewable energy sources (Fuelwood and LPG)

Emission reduction due to the consumption of non-renewable energy sources has been accounted in accordance with the GS TPDDTEC 3.1 methodology using the following equation:

Where:

$$BE_{b,y} = B_{b,y} * ((f_{NRB,y} * EF_{b,fuel, CO2}) + EF_{b,fuel, nonCO2}) * NCV_{b, fuel}$$

Equation 3

$B_{b,y}$	= Quantity of fuel consumed in baseline scenario b during year y, in tons,
f_{nrb}	= Fraction of biomass used during year y for the considered scenario that can be established as non-renewable biomass
$NCV_{b,fuel}$	= Net calorific value of the fuel that is substituted or reduced (IPCC default for wood fuel, 0.015 TJ/ton)
$Ef_{b,fuel,CO2}$	CO ₂ emission factor of the fuel that is substituted or reduced. 112 tCO ₂ /TJ for Wood/Wood Waste, or the IPCC default value of other relevant fuel
$Ef_{b,fuel,non-CO2}$	= Non-CO ₂ emission factor of the fuel that is substituted or reduced

Project Emissions (PE_y)

As per GS TPDDTEC 3.1, project emissions due to project activity will be accounted from:

(i) Project emissions due to continued use of baseline technology

These will be calculated using the following equation:

$$PE_{p,y} = B_{p,y} * ((f_{NRB,y} * EF_{p,fuel, CO2}) + EF_{p,fuel, nonCO2}) * NCV_{p, fuel}$$

Equation 4

Where:

$PE_{p,y}$	= Emissions for project scenario p during year y in tCO ₂ e
$B_{p,y}$	= Quantity of fuel consumed in project scenario p during year y, in tons
f_{nrb}	= Fraction of biomass used during year y for the considered scenario that can be established as non-renewable biomass
$NCV_{p,fuel}$	= Net calorific value of the fuel that is substituted or reduced (IPCC default for wood fuel, 0.015 TJ/ton)
$Ef_{p,fuel,CO2}$	= CO ₂ emission factor of the fuel that is substituted or reduced. 112 tCO ₂ /TJ for Wood/Wood Waste, or the IPCC default value of other relevant fuel

$E_{p, \text{fuel, non-CO}_2}$ = Non-CO₂ emission factor of the fuel that is substituted or reduced

(ii) Project emissions from biodigesters

These include physical leakage and incomplete combustion of biogas, as well as emissions from the animal waste not treated in the bio-digester

The first two components are calculated as a percentage of the methane produced, as per the following equation:

$$PE_{\text{awms}, h, y} = GWP_{\text{CH}_4} * \sum (N_{(T), h, y} \cdot EF_{\text{awms}, T}) \cdot PL_y + \sum (N_{(T), h, y} \cdot EF_{\text{awms}, T}) \cdot (1 - \eta_{\text{biogastove}})(1 - PL_y)$$

Where:

$N_{(T), h, y}$	= Number of animals of livestock category T in year y in premise h
$EF_{\text{awms}, T}$	= CH ₄ emission factor for livestock category T (tCH ₄ per animal per year)
PL_y	= Default value of 10%
GWP_{CH_4}	= Global Warming Potential of Methane, 28
$\eta_{\text{biogastove}}$	= Combustion efficiency of the used type of biogas stove to account for incomplete combustion resulting in emission of methane post combustion.

$$EF_{awms(T)} = VS_{(T)} * 365 * \left[Bo_{(T)} * D_{CH_4} * \sum_k \frac{MCF_{BL,k}}{100} * MS_{(T,k)} \right] \tag{16}$$

Where:

- EF_{awms(T)} CH₄ emission factor for livestock category *T*, (tCH₄per animal per year)
- VS_(T) Daily volatile solid excreted for livestock category *T*, (kg dry matter per animal per day)
- 365 Basis for calculating annual VS production, (days per year)
- Bo_(T) Maximum methane production capacity for manure produced by livestock category *T*, (m³CH₄ per kg of VS excreted)
- D_{CH₄} CH₄ density (0.00067 t per m³ at room temperature (20 °C) and 1 atm pressure)
- MCF_(BL,k) Methane conversion factors for the animal waste handling system in the baseline situation by climate zone *k*, (%)
- MS_(T,S,k) Fraction of livestock category *T*'s manure treated in the animal waste management system, in climate region *k* (dimensionless)

We will estimate methane emissions from untreated in accordance with equation 1, 2 and 16 above with the following changed parameters:

MCF_(P,S,k) Methane conversion factors for the animal waste handling system used in addition to bio-digester in the project scenario by climate zone *k*, (%)

MS_(P,S,k) Fraction of livestock category *T*'s manure not treated in bio-digester, in climate region *k*, (dimensionless)

VS_(T) Daily volatile solid excreted for livestock category *T*, (kg dry matter per animal per day)

Bo_(T) Maximum methane production capacity for manure produced by livestock category *T*, (m³CH₄ per kg of VS excreted)

Leakage (LEy): The potential sources of leakages are discussed in details in section B.7.3 below.

Year	Total ERs (tCO2e)
1	42,217

2	56,512
3	65,817
4	65,817
5	65,817
TOTAL	296,181
Annual average	59,236

Methodological action to estimate SDG 3 outcomes: Beneficiaries perceiving improvement in indoor air quality AND Beneficiaries reporting better hygiene conditions. There is no specific methodology to determine the impact to this indicator. The PD will apply periodical surveys.

Methodological action to estimate SDG 5 outcomes: There is no specific methodology to determine the impact to this indicator. The PD will apply periodical surveys and field test to collect data for this SDG.

Selected parameters: Women with access to technology, and reported time saved due to use of biogas, and share of women participating in the project

Methodological action to estimate SDG 7 outcomes:

Indicator: 7.1.2 Proportion of population with primary reliance on clean fuels and technology

Selected parameter: Number of connections of clean and renewable energy source AND Number of persons getting access to a clean and renewable energy source

Approach for estimation of the outcome: The renewable thermal energy generated with biogas will be calculated based on unit in use, amount of biogas used and stoves efficiency.

Methodological action to estimate SDG 8 outcomes:

Indicator:

8.5.1 Average hourly earnings of female and male employees, by occupation, age and persons with disabilities

8.5.2 Unemployment rate, by sex, age and persons with disabilities

8.6.1 Proportion of youth (aged 15-24 years) not in education, employment or training

Selected parameter: Number of permanent jobs generated and number of staff trained

Approach for estimation of the outcome: No specific calculations. Number of jobs generated every year due to the implementation of project activity and trainings offered to project staff will be monitored based on the HR records of Buen Manejo del Campo S.A de C.V through BambooHR software.

Methodological action to estimate SDG 15 outcomes:

Indicator:

15.3.1 Proportion of land that is degraded over total land area

Approach for estimation of the outcome: The recommended amount of bio-fertilizer to use is 10m³ per hectare. Considering capacity of the different reactor systems and retention time, it is estimated that production of bio-fertilizer per year.

B.6.2. Data and parameters fixed ex ante

SDG13 Climate Action

Data/parameter	VS _(T)					
Unit	kg dry matter per animal per day (kg/hd/day)					
Description	Daily volatile solid excreted for livestock category T					
Source of data	2006 IPCC Guidelines for National Greenhouse Gas Inventories					
Value(s) applied		Dairy, liquid with cover	Dairy, Liquid Storage	Dairy, lagoon storage	Swine, liquid storage with crust	Swine, anaerobic lagoon storage
	VS _(T) North America	5.4	5.4	5.4	0.27	0.27
Choice of data or Measurement methods and procedures	Deemed valid by GS VER Methodology					
Purpose of data	Calculation of baseline and project emissions. Determination of CH ₄ emissions.					
Additional comment	365 = basis for calculating annual VS production, days per year. North America values has been chosen for the following reasons: 1) Geographically, Mexico is considered part of North America region; 2) The National Institute of					

	Climate Change (INECC for its meaning in Spanish) considers appropriate to calculate the country emission from agricultural and livestock sector applying the value for North America from the IPCC. Evidence of that can be found in the Tool "Mayahuel" developed by the INECC that was used to calculate the national inventory of GHG. The tool is available in the following link (opened on 14 May 2021): https://www.gob.mx/inecc/documentos/investigaciones-2018-2013-en-materia-de-mitigacion-del-cambio-climatico. See tab 'M1-IOB (Vaca lechera-LB)', cells C72:C77.
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Data / Parameter	MCF _(BL,k)					
Unit	%					
Description	Methane conversion factors for the animal waste handling system in the baseline situation by climate zone k					
Source of data	2006 IPCC Guidelines for National Greenhouse Gas Inventories					
Value(s) applied		Dairy, liquid with cover	Dairy, Liquid Storage	Dairy, lagoon storage	Swine, liquid storage with crust	Swine, anaerobic lagoon storage
	MCF _(BL,k) Temperate	26%	42%	78%	26%	78%
	MCF _(BL,k) Warm	44%	71%	79%	44%	79%
Choice of data or Measurement methods and procedures	Deemed valid by GS VER Methodology					
Purpose of data	Calculation of baseline and project emissions. Determination of CH ₄ emissions.					
Additional comment	North America values has been chosen for the following reasons: 1) Geographically, Mexico is considered part of North America region; 2) The National Institute of Climate Change (INECC for its meaning in Spanish) considers appropriate to calculate the country emission from agricultural and livestock sector applying the value					

	for North America from the IPCC. Evidence of that can be found in the Tool "Mayahuel" developed by the INECC that was used to calculate the national inventory of GHG. The tool is available in the following link (opened on 14 May 2021): https://www.gob.mx/inecc/documentos/investigaciones-2018-2013-en-materia-de-mitigacion-del-cambio-climatico. See tab 'M1-IOB (Vaca lechera-LB)', cells C72:C77.
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Data / Parameter	Bo(T)					
Unit	m ³ CH ₄ /kg VS					
Description	Maximum methane production capacity for manure produced by livestock category T, (m ³ CH ₄ per kg of VS excreted)					
Source of data	2006 IPCC Guidelines for National Greenhouse Gas Inventories					
Value(s) applied		Dairy, liquid with cover	Dairy, Liquid Storage	Dairy, lagoon storage	Swine, liquid storage with crust	Swine, anaerobic lagoon storage
	Bo(T)	0.24	0.24	0.24	0.48	0.48
Choice of data or Measurement methods and procedures	Deemed valid by GS VER Methodology					
Purpose of data	Calculation of baseline and project emissions. Determination of CH ₄ emissions.					
Additional comment	North America values has been chosen for the following reasons: 1) Geographically, Mexico is considered part of North America region; 2) The National Institute of Climate Change (INECC for its meaning in Spanish) considers appropriate to calculate the country emission from agricultural and livestock sector applying the value for North America from the IPCC. Evidence of that can be found in the Tool "Mayahuel" developed by the INECC that was used to calculate the national inventory of GHG. The tool is available in the following link (opened on 14 May 2021):					

	https://www.gob.mx/inecc/documentos/investigaciones-2018-2013-en-materia-de-mitigacion-del-cambio-climatico . See tab 'M1-IOB (Vaca lechera-LB)', cells C72:C77.
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Data / Parameter	D_{CH4}
Unit	kg per m ³
Description	Methane (CH ₄) density (0.00067 t per m ³ at room temperature (20 °C) and 1 atm pressure)
Source of data	GS VER Methodology
Value(s) applied	0.67
Choice of data or Measurement methods and procedures	It is considered reasonable based on the project proponent experience working with small farmers.
Purpose of data	Calculation of baseline and project emissions. Determination of CH ₄ emissions.
Additional comment	

Data / Parameter	f_{NRB}
Unit	%
Description	Non-renewability status of woody biomass fuel in scenario i during year y
Source of data	f _{NRB} Calculation
Value(s) applied	0.8974
Choice of data or Measurement methods and procedures	Value calculated using methodological approach justified with relevant and updated data.
Purpose of data	Determination of the emission reductions of the project activity as per equation 1 above.
Additional comment	

Data/parameter	EF_{b,co2}
Unit	tco ₂ /TJ
Description	Co ₂ emission factor arising from use of fuels in baseline scenario (wood fuel is considered as the baseline fuel.)
Source of data	IPCC default values IPCC default value IPCC 2006 Guidelines for National Greenhouse gas Inventories Chapter 2: Stationary Combustion (https://www.ipcc-nggip.iges.or.jp/public/2006gl/pdf/2_Volume2/V2_2_Ch2_Stationary_Combustion.pdf) Page 2.23/ Table 2.5
Value(s) applied	112
Choice of data or Measurement methods and procedures	Deemed valid by GS VER Methodology
Purpose of data	Determination of CO ₂ emission factor in baseline
Additional comment	

Data/parameter	EF_{b,non co2}
Unit	tco ₂ /TJ
Description	Non-co ₂ emission factor arising from use of fuels in baseline scenario
Source of data	GWP: IPCC AR5, https://www.ipcc.ch/assessment-report/ar5/ CH ₄ and N ₂ O Emission Factors: Emission Factor value provided in Table 2.5 of Chapter 2: Stationary Emissions

	(2006 IPCC Guidelines for National Greenhouse Gas Inventories). https://www.ipcc-nggip.iges.or.jp/public/2006gl/pdf/2_Volume2/V2_2_Ch2_Stationary_Combustion.pdf
Value(s) applied	9.46 (value applied from 01/10/2020 onwards)
Choice of data or Measurement methods and procedures	(CH ₄ =0.3*GWP 28) + (N ₂ O=0.004*GWP 265) = 9.46 Deemed valid by GS VER Methodology Determined as per IPCC default figures
Purpose of data	Determination of non-CO ₂ emission factor in baseline
Additional comment	

Data/parameter	EF_{p,co2}
Unit	tco ₂ /TJ
Description	co ₂ emission factor arising from use of fuels in project scenario
Source of data	IPCC default values IPCC 2006 Guidelines for National Greenhouse gas Inventories Chapter 2: Stationary Combustion (https://www.ipcc-nggip.iges.or.jp/public/2006gl/pdf/2_Volume2/V2_2_Ch2_Stationary_Combustion.pdf) Page 2.23/ Table 2.5
Value(s) applied	112
Choice of data or Measurement methods and procedures	Deemed valid by Methodology Determined as per IPCC default figures

Purpose of data	Determination of CO2 emission factor in project
Additional comment	

Data/parameter	EF_{p,non co2}
Unit	tco ₂ /TJ
Description	Non-co ₂ emission factor arising from use of fuels in project scenario
Source of data	GWP: IPCC AR5, https://www.ipcc.ch/assessment-report/ar5/ CH ₄ and N ₂ O Emission Factors: Emission Factor value provided in Table 2.5 of Chapter 2: Stationary Emissions (2006 IPCC Guidelines for National Greenhouse Gas Inventories). https://www.ipcc-nggip.iges.or.jp/public/2006gl/pdf/2_Volume2/V2_2_Ch2_Stationary_Combustion.pdf
Value(s) applied	9.46
Choice of data or Measurement methods and procedures	(CH ₄ =0.3*GWP 28) + (N ₂ O=0.004*GWP 265) Deemed valid by GS VER Methodology Determined as per IPCC default figures
Purpose of data	Determination of non-CO ₂ emission factor in project
Additional comment	

Data/parameter	NCV_b
Unit	TJ/ton
Description	Net calorific value of the fuels used in the baseline (Wood and LPG)
Source of data	LPG: IPCC default value

	IPCC (2006) "IPCC Guidelines for National Greenhouse Gas Inventories", Volume 2, Energy, Chapter 1, Introduction, Table 1.2, p 1.18 Wood: Energy National Balance 2019 (page 95) https://www.gob.mx/sener/documentos/balance-nacional-de-energia-2019
Value(s) applied	Wood: 0.014486 LPG: 0.0473
Choice of data or Measurement methods and procedures	Deemed valid by Methodology Determined as per IPCC default figures and official local source
Purpose of data	Determination of fuel's NCV in baseline
Additional comment	

Data/parameter	NCV_p
Unit	TJ/ton
Description	Net calorific value of the fuels used in the project (Wood and LPG)
Source of data	LPG: IPCC default value IPCC (2006) "IPCC Guidelines for National Greenhouse Gas Inventories", Volume 2, Energy, Chapter 1, Introduction, Table 1.2, p 1.18 Wood: Energy National Balance 2019 (page 95) https://www.gob.mx/sener/documentos/balance-nacional-de-energia-2019
Value(s) applied	Wood: 0.014486

	LPG: 0.0473
Choice of data or Measurement methods and procedures	Deemed valid by Methodology Determined as per IPCC default figures and official local source
Purpose of data	Determination of fuels NCV in project
Additional comment	

SDG3 Good health and wellbeing

The use of biogas stove reduces harmful indoor air pollution emissions produced when burning fuelwood. The soot and ashes inside the homes when using the traditional open fire affects the cleanliness. The black walls and ceiling prevent people feeling a clean and neat space.

The use of the biodigester reduces presence of diseases vectors (e.g. flies and similar insects), and reduces odor, therefore the hygiene conditions are improved.

SDG5 Gender equality

- Through experience, focus groups, and targeted surveys, we have found that women save time through the use of Sistema.bio's technology, specifically by making obsolete the collecting of fuel (or reducing the amount needed), reduce the time needed to attending fires and cleaning pots. We will measure reported time savings due to biogas through a questionnaire that will identify the number of hours per month that users have saved due to the use of the technology, this data will be disaggregated by gender. The project will also measure the number of women with access to the technology and the proportion of women working in the project.

SDG7 Affordable and clean energy

The project provides access to clean, renewable and reliable energy through the use of the biodigester technology.

SDG15 Life on Land

The biodigester technology produces organic biofertilizer that is used on fields and crops and has a positive impact on the quality of soil and plants. Farmers receive

training on the use of biofertilizer in their farms to support the transition to organic farming.

B.6.3. Ex ante estimation of SDG Impact

SDG Goal	Baseline Situation	Expected impact
SDG 13 – Take urgent action to combat climate change and its impacts Target: 13.2, 13.3 Indicator: 13.2.1, 13.3.1	Baseline estimated emissions, annual average over the crediting period: 72,331 tCO ₂ e	Net benefit, estimated emissions, annual average over the crediting period: 59,236 tCO ₂ e
SDG3 - No Poverty Target 3.1 By 2030, substantially reduce the number of deaths and illnesses from hazardous chemicals and air, water and soil pollution and contamination Target: 3.9 By 2030, substantially reduce the number of deaths and illnesses from hazardous chemicals and air, water and soil pollution and contamination Indicator 3.9.1 Mortality rate attributed to household and ambient air pollution	The soot and ashes inside the homes when using the traditional open fire affects the cleanliness. The black walls and ceiling prevent people feeling a clean and neat space. Traditional manure management causes presence of diseases vectors like flies and bad odor.	90% Beneficiaries reporting air inside their homes is cleaner AND 90% Beneficiaries reporting better hygiene conditions
SDG 5 – Gender equality Target: 5.4 Indicator 5.4.1 Proportion of time spent on unpaid domestic and care work, by sex, age and location	– No women with access to technology – Zero hour saved due to use of biogas – Share of women participating in the project	– 2,800 women – 8.4 hours saved per week due to use of biogas – 50% women participating in the project
SDG 7 – Affordable and clean Energy Target: 7.1, 7.2 Indicator: 7.1.2 Proportion of source AND population with primary	Number of connections of clean and renewable energy	3,000 Number of connections 14,400. Number of persons

reliance on clean fuels and technology	Number of persons getting access to a clean and renewable energy source	
8 Decent work and economic growth		
Target: 8.5.1 Average hourly earnings of female and male employees, by occupation, age and persons with disabilities	Zero permanent jobs created	20 permanent jobs created
8.5.2 Unemployment rate, by sex, age and persons with disabilities	Zero training offered to staff	40 trainings for project staff
8.6.1 Proportion of youth (aged 15-24 years) not in education, employment or training		
15 Life on land		
Target 15.3: By 2030, combat desertification, restore degraded land and soil, including land affected by desertification, drought and floods, and strive to achieve a land degradation-neutral world	Zero hectares fertilized with Bio-fertilizer	7,537 hectares fertilized with Bio-fertilizer
Indicator:	Final use of bio-fertilizer: No bio-fertilizer being used.	Bio-fertilizer being sold; used by the beneficiary; or given free of charge
15.3.1 Proportion of land that is degraded over total land area		

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B.6.4. Summary of ex ante estimates of each SDG Impact

SDG13 Climate Action

Year	Baseline estimate	Project estimate	Net benefit
Year 1	50,212	7,995	42,217
Year 2	76,335	12,126	56,512

Year 2	78,369	12,512	65,817
Year 4	78,369	12,512	65,817
Year 5	78,369	12,512	65,817
Total	361,655	57,618	294,181
Total number of crediting years	5		
Annual average over the crediting period	72,331	11,524	59,236 ³⁶

SDG3 Good health and wellbeing

The PD will carry out periodic surveys to define this parameter.

SDG 5 Gender equality

The PD will carry out periodic surveys to define this parameter.

SDG 7 Affordable and clean energy

The PD will carry out periodic surveys to define this parameter.

SDG 8 Decent Work and Economic Growth

The project will report the contract and payroll evidence.

SDG 15 Life on Land

The project will consider capacity of the different reactor systems and retention time, to estimate the production of bio-fertilizer per year.

B.7. Monitoring plan

B.7.1. Data and parameters to be monitored

SDG 13

Data/parameter	$P_{b, \text{firewood}, y}$
Unit	Tonnes firewood per household per year
Description	Quantity of firewood consumed in baseline scenario b during year y
Source of data	Monitoring survey

³⁶ The Emission Reduction includes 40 toCO₂e per year of leakage. Leakage is not shown in the table.

Value(s) applied	0.01039 tons/household/day (ex-ante value, this value to be updated based on monitoring).
Measurement methods and procedures	Proxy field test result
Monitoring frequency	Every two years (or more frequently), reported annually.
QA/QC procedures	Continuous supervision on monitoring campaigns and crosscheck of results and consolidation of databases and results. QA/QC to take place at operational and administrative levels including checks by external (e.g. project consultant).
Purpose of data	Estimation of CO ₂ e emission reductions
Additional comment	Calculated based on 0.79 t/capita/yr * 4.8 Persons per HH / 365 days

Data/parameter	P_{b,LPG,y}
Unit	Tonnes LPG per household per year
Description	Quantity of LPG consumed in baseline scenario b during year y
Source of data	Monitoring survey
Value(s) applied	0.00097 tons/household/day (ex-ante value, this value to be updated based on monitoring).
Measurement methods and procedures	Proxy field test result
Monitoring frequency	Every two years (or more frequently), reported annually.
QA/QC procedures	Continuous supervision on monitoring campaigns and crosscheck of results and consolidation of databases and results. QA/QC to take place at operational and administrative levels including checks by external (e.g. project consultant).
Purpose of data	Estimation of CO ₂ e emission reductions

Additional comment	0.00097 (t/hh/yr) Calculated based on: 0.203 kg/capita/day * 4.8 Persons per HH / 1000
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Data / Parameter	P_p firewood,y
Unit	Tonnes firewood per household per year
Description	Quantity of firewood consumed in baseline scenario b during year y
Source of data	Monitoring survey
Value(s) applied	0.001039 tons/household/day (ex-ante value, this value to be updated based on monitoring).
Measurement methods and procedures	Proxy field test result
Monitoring frequency	Updated every two years, or more frequently
QA/QC procedures	Continuous supervision on monitoring campaigns and crosscheck of results and consolidation of databases and results. QA/AC to take place at operational and administrative levels including checks by external (e.g. project consultant).
Purpose of data	Estimation of CO ₂ e emission reductions
Additional comment	0.001039 tons/household/day Calculated based on: (0.79 t/capita/yr * 4.8 Persons per HH / 365 days)* .10 For the ex-ante calculation, a value was calculated based on the premise that 10% of the firewood and LPG are still used in the project scenario.

Data / Parameter	P_p LPG,y
Unit	Tonnes LPG per household per year
Description	Quantity of LPG consumed in baseline scenario b during year y

Source of data	Monitoring survey
Value(s) applied	0.000097 tons/household/day (ex-ante value, this value to be updated based on monitoring).
Measurement methods and procedures	Proxy field test result
Monitoring frequency	Every two years (or more frequently), reported annually.
QA/QC procedures	Continuous supervision on monitoring campaigns and crosscheck of results and consolidation of databases and results. QA/QC to take place at operational and administrative levels including checks by external (e.g. project consultant).
Purpose of data	Estimation of CO ₂ e emission reductions
Additional comment	0.000097 (t/hh/yr) Calculated based on: (0.203 kg/capita/day * 4.8 Persons per HH / 1000) *.10 % For the ex-ante calculation, a value was calculated based on the premise that 10% of the firewood and LGP are still used in the project scenario.

Data / Parameter	N_{(T)h}																																		
Unit	Numeric																																		
Description	The number of animals of livestock species per category T																																		
Source of data	Baseline and monitoring survey																																		
Value(s) applied	<table><tr><th>Sistema Size</th><th>Dairy cows (Warm zone)</th><th>Swine (Warm zone)</th><th>Dairy cows (Temperate zone)</th><th>Swine (Temperate zone)</th></tr><tr><td>Sistema 6</td><td>5</td><td>19</td><td>4</td><td>16</td></tr><tr><td>Sistema 8</td><td>7</td><td>25</td><td>5</td><td>22</td></tr><tr><td>Sistema 12</td><td>9</td><td>38</td><td>7</td><td>28</td></tr><tr><td>Sistema 16</td><td>13</td><td>50</td><td>10</td><td>41</td></tr><tr><td>Sistema 20</td><td>18</td><td>75</td><td>14</td><td>56</td></tr></table>					Sistema Size	Dairy cows (Warm zone)	Swine (Warm zone)	Dairy cows (Temperate zone)	Swine (Temperate zone)	Sistema 6	5	19	4	16	Sistema 8	7	25	5	22	Sistema 12	9	38	7	28	Sistema 16	13	50	10	41	Sistema 20	18	75	14	56
Sistema Size	Dairy cows (Warm zone)	Swine (Warm zone)	Dairy cows (Temperate zone)	Swine (Temperate zone)																															
Sistema 6	5	19	4	16																															
Sistema 8	7	25	5	22																															
Sistema 12	9	38	7	28																															
Sistema 16	13	50	10	41																															
Sistema 20	18	75	14	56																															

	Sistema 30	26	113	20	81
	Sistema 40	35	147	27	109
Measurement methods and procedures	Field monitoring survey				
Monitoring frequency	Every two years, reported annually.				
QA/QC procedures	Continuous supervision on monitoring campaigns and crosscheck of results and consolidation of databases and results. QA/QC to take place at operational and administrative levels including checks by external (e.g. project consultant).				
Purpose of data	Estimation of CO ₂ e emission reductions				
Additional comment					

Data / Parameter	$U_{p,y}$
Unit	Percentage
Description	Usage rate in project scenario p during year y
Source of data	Annual usage survey/Monitoring survey
Value(s) applied	Ex ante estimate: 90%
Measurement methods and procedures	The usage survey is carried out annually.
Monitoring frequency	Annual
QA/QC procedures	Continuous supervision on monitoring campaigns and crosscheck of results and consolidation of databases and results. QA/QC to take place at operational and administrative levels including checks by external (e.g. project consultant).
Purpose of data	Estimation of CO ₂ e emission reductions
Additional comment	A usage parameter is derived for biodigesters and biogas stoves being credited. The project follows the Rule update: Requirements and Guidelines for carrying out usage surveys for projects implementing improved cooking devices, dated

23/08/2017, and adheres to the usage guidelines and the rule update made public on 27/10/2020.
See below further information about definitions of “use and “non-use” for both, the biodigester and the biogas stove.

Data / Parameter	N _{p,y}																				
Unit	Number of project technologies-days credited																				
Description	Cumulative number of project technology-days included in the project database for project scenario p against baseline scenario b in year y																				
Source of data	Total sales and distribution record																				
Value(s) applied	Ex ante estimation: <table><tr><th>Year</th><th>Biodigesters</th><th>Biogas stoves</th></tr><tr><td>1</td><td>547,500</td><td>547,500</td></tr><tr><td>2</td><td>1,095,000</td><td>1,095,000</td></tr><tr><td>3</td><td>1,095,000</td><td>1,095,000</td></tr><tr><td>4</td><td>1,095,000</td><td>1,095,000</td></tr><tr><td>5</td><td>1,095,000</td><td>1,095,000</td></tr></table>			Year	Biodigesters	Biogas stoves	1	547,500	547,500	2	1,095,000	1,095,000	3	1,095,000	1,095,000	4	1,095,000	1,095,000	5	1,095,000	1,095,000
Year	Biodigesters	Biogas stoves																			
1	547,500	547,500																			
2	1,095,000	1,095,000																			
3	1,095,000	1,095,000																			
4	1,095,000	1,095,000																			
5	1,095,000	1,095,000																			
Measurement methods and procedures	Continuous monitoring. The number of biodigesters and biogas stoves is recorded in the selling database.																				
Monitoring frequency	Continuous																				
QA/QC procedures	Continuous supervision on monitoring campaigns and crosscheck of results and consolidation of databases and results. QA/AC to take place at operational and administrative levels including checks by external (e.g. project consultant).																				
Purpose of data	Estimation of CO ₂ e emission reductions																				
Additional comment	The operational and maintenance days will be tracked. The days when the biodigester is not operating will be discounted from N.p.y parameter.																				

Transparent records will be made available to demonstrate how discounted days due to maintenance are considered in the calculations.

Data / Parameter	EF _{awms,T}																											
Unit	kgCH4per animal per year																											
Description	CH4 emission factor for livestock category T,																											
Source of data	The emission factors are calculated using the data in Tables 10A-4 through 10A-9 from the 2006 IPCC Guidelines for National Greenhouse Gas Inventories. Default methane conversion factors (MCFs) are provided in Table 10.17 from the 2006 IPCC Guidelines for National Greenhouse Gas Inventories for different manure management systems and by annual average temperatures.																											
Value(s) applied	<table><tr><td></td><td>Dairy Liquid slurry without natural crust cover</td><td>Dairy, Liquid/Slurry (With natural crust cover)</td><td>Dairy, Uncovered anaerobic lagoon</td><td>Swine, Liquid slurry without natural crust cover</td><td>Swine, Liquid/Slurry (With natural crust cover)</td><td>Swine, Uncovered anaerobic lagoon</td></tr><tr><td>EF_{awms(T)}</td><td>11.66</td><td>66.18</td><td>197.77</td><td>1.17</td><td>6.62</td><td>19.78</td></tr><tr><td>kg</td><td>18.00</td><td>104.97</td><td>200.30</td><td>1.80</td><td>10.50</td><td>20.03</td></tr></table>								Dairy Liquid slurry without natural crust cover	Dairy, Liquid/Slurry (With natural crust cover)	Dairy, Uncovered anaerobic lagoon	Swine, Liquid slurry without natural crust cover	Swine, Liquid/Slurry (With natural crust cover)	Swine, Uncovered anaerobic lagoon	EF _{awms(T)}	11.66	66.18	197.77	1.17	6.62	19.78	kg	18.00	104.97	200.30	1.80	10.50	20.03
	Dairy Liquid slurry without natural crust cover	Dairy, Liquid/Slurry (With natural crust cover)	Dairy, Uncovered anaerobic lagoon	Swine, Liquid slurry without natural crust cover	Swine, Liquid/Slurry (With natural crust cover)	Swine, Uncovered anaerobic lagoon																						
EF _{awms(T)}	11.66	66.18	197.77	1.17	6.62	19.78																						
kg	18.00	104.97	200.30	1.80	10.50	20.03																						
Measurement methods and procedures	Calculated using the equation 2 of this PDD.																											
Monitoring frequency	Annually																											
QA/QC procedures	The average MS% for each size of biodigester would be taken into consideration for calculation and will vary upon verification.																											
Purpose of data	Baseline emissions																											
Additional comment	NA																											
	NA																											

Data / Parameter	LE_{p,y}
Unit	t_CO ₂ e per year
Description	Leakage in project scenario p during year y
Source of data	Baseline and monitoring survey
Value(s) applied	40
Measurement methods and procedures	Field monitoring survey
Monitoring frequency	Every two years
QA/QC procedures	Continuous supervision on monitoring campaigns and crosscheck of results and consolidation of databases and results. QA/QC to take place at operational and administrative levels including checks by external (e.g. project consultant).
Purpose of data	Estimation of CO ₂ e emission reductions
Additional comment	The potential sources of leakages will be assessed at the verification. More information on the potential source of leakages can be found below in section B.7.3

SDG 3

Data / Parameter	% Beneficiaries reporting air inside the home is cleaner
Unit	%
Description	Households are surveyed to determine if they report the air is cleaner after installation of the biogas stove.
Source of data	Responses to qualitative surveys (based on results from monitoring surveys collected)
Value(s) applied	90%
Measurement methods and procedures	Using smartphone devices, supervisors collect surveys which are stored and reported in a Salesforce.com database. Randomness of sample maintained by surveying every nth client who receives a supervisory visit from Sistema.bio.
Monitoring frequency	Annually

QA/QC procedures	Continuous supervision on monitoring campaigns and crosscheck of results and consolidation of databases and results. QA/QC to take place at operational and administrative levels including checks by external (e.g. project consultant).
Purpose of data	Assess sustainability
Additional comment	

Data / Parameter	% Beneficiaries reporting better hygiene conditions
Unit	%
Description	Households are surveyed to determine if they report the hygiene conditions have improved after the installation of the biodigester.
Source of data	Responses to qualitative surveys (based on results from monitoring surveys collected)
Value(s) applied	90%
Measurement methods and procedures	Using smartphone devices, supervisors collect surveys which are stored and reported in a Salesforce.com database. Randomness of sample maintained by surveying every nth client who receives a supervisory visit from Sistema.bio.
Monitoring frequency	Annually
QA/QC procedures	Continuous supervision on monitoring campaigns and crosscheck of results and consolidation of databases and results. QA/QC to take place at operational and administrative levels including checks by external (e.g. project consultant).
Purpose of data	Assess sustainability
Additional comment	

Data / Parameter	Country or province/state where project is located
Unit	Country or province/state name
Description	Project location
Source of data	Country or province/state where the project is located
Value(s) applied	Mexico

Measurement methods and procedures	Continuous update of the project database
Monitoring frequency	Single monitoring campaign
QA/QC procedures	Continuous supervision on monitoring campaigns and crosscheck of results and consolidation of results. QA/QC to take place at operational and administrative levels including checks by external (e.g. project consultant).
Purpose of data	Quantification of impact relating SDG3
Additional comment	

SDG 5

Data / Parameter	Women with access to technology
Unit	Number of women
Description	Women having access to clean and renewable energy technology for cooking.
Source of data	Responses to monitoring surveys (based on results from monitoring surveys collected)
Value(s) applied	2,800
Measurement methods and procedures	Using smartphone devices, supervisors collect surveys which are stored and reported in a Salesforce.com database. Randomness of sample maintained by surveying every nth client who receives a supervisory visit from Sistema.bio.
Monitoring frequency	Annually
QA/QC procedures	Continuous supervision on monitoring campaigns and crosscheck of results and consolidation of databases and results. QA/QC to take place at operational and administrative levels including checks by external (e.g. project consultant).
Purpose of data	Assess sustainability
Additional comment	

Data / Parameter	Hours saved due to use of biogas
Unit	Hours per week
Description	The ready to use fuel can save time dedicated to collect fuelwood. The thermal power of the biogas help to reduce time of cooking must of meals prepared.
Source of data	Responses to monitoring surveys (based on results from monitoring surveys collected)
Value(s) applied	8.4
Measurement methods and procedures	Using smartphone devices, supervisors collect surveys which are stored and reported in a Salesforce.com database. Randomness of sample maintained by surveying every nth client who receives a supervisory visit from Sistema.bio.
Monitoring frequency	Annually
QA/QC procedures	Continuous supervision on monitoring campaigns and crosscheck of results and consolidation of databases and results. QA/QC to take place at operational and administrative levels including checks by external (e.g. project consultant).
Purpose of data	Assess sustainability
Additional comment	

Data / Parameter	Share of women participating in the project
Unit	%
Description	The project will report the share of women employed.
Source of data	Human resources records.
Value(s) applied	50
Measurement methods and procedures	Internal records including Contracts
Monitoring frequency	Annually

QA/QC procedures	Human Resources department uses the application BambooHR to store evidences and will ensure veracity and accuracy of the information.
Purpose of data	Assess sustainability
Additional comment	The monitoring frequency will be annual but in case an employee hired works for less than a year, it will be reported but including a note.

SDG 7

Data / Parameter	Number of connections of clean and renewable energy source
Unit	Number of connections
Description	The project will count how many connections of clean and renewable energy have been made due the installation of the biodigesters and biogas stoves.
Source of data	Sales records
Value(s) applied	3,000
Measurement methods and procedures	Continuous monitoring. The number of biodigesters and biogas stoves is recorded in the selling database.
Monitoring frequency	Continuous
QA/QC procedures	Continuous supervision on monitoring campaigns and crosscheck of results and consolidation of databases and results. QA/AC to take place at operational and administrative levels including checks by external (e.g. project consultant).
Purpose of data	Assess sustainability
Additional comment	

Data / Parameter	Number of persons getting access to a clean and renewable energy source
Unit	Number of persons

Description	The project will count number of units installed times the average number of persons at the household.
Source of data	Sales records and monitoring surveys and Responses to qualitative surveys (based on results from monitoring surveys collected)
Value(s) applied	14,400
Measurement methods and procedures	Continuous monitoring. The number of biodigesters and biogas stoves is recorded in the selling database. Using smartphone devices, supervisors collect surveys which are stored and reported in a Salesforce.com database. Randomness of sample maintained by surveying every nth client who receives a supervisory visit from Sistema.bio
Monitoring frequency	Number of units continually Average number of persons at the household annually.
QA/QC procedures	Continuous supervision on monitoring campaigns and crosscheck of results and consolidation of databases and results. QA/QC to take place at operational and administrative levels including checks by external (e.g. project consultant). Continuous supervision on monitoring campaigns and crosscheck of results and consolidation of databases and results. QA/QC to take place at operational and administrative levels including checks by external (e.g. project consultant).
Purpose of data	Assess sustainability
Additional comment	

SDG 15

Data / Parameter	Final use of bio-fertilizer
Unit	NA
Description	Bio-fertilizer being sold, used by the beneficiary, or given free of charge

Source of data	Monitoring Survey
Value(s) applied	To be monitored
Measurement methods and procedures	The usage survey is carried out annually.
Monitoring frequency	Annual
QA/QC procedures	Continuous supervision on monitoring campaigns and crosscheck of results and consolidation of databases and results. QA/QC to take place at operational and administrative levels including checks by external (e.g. project consultant).
Purpose of data	Determine the final use of the by-product generated at the biodigester.
Additional comment	This paramant has been included in response to a comment received at the preliminary review, but also the use of a biofertilizer can be considered a positive impact.

Data / Parameter	Hectares fertilized with Bio-fertilizer
Unit	NA
Description	Bio-fertilizer being sold, used by the beneficiary, or given free of charge
Source of data	Monitoring Survey
Value(s) applied	7,537
Measurement methods and procedures	The usage survey is carried out annually.
Monitoring frequency	Annual
QA/QC procedures	Continuous supervision on monitoring campaigns and crosscheck of results and consolidation of databases and results. QA/QC to take place at operational and administrative levels including checks by external (e.g. project consultant).
Purpose of data	Determine the final use of the by-product generated at the biodigester.
Additional comment	This paramant has been included in response to a comment received at the preliminary review, but also the use of a biofertilizer can be considered a positive impact.

SDG 8

Data / Parameter	Training offered to staff
Unit	Number and type of training
Description	All training offered to project staff will be monitored and reported including date, list of participants, duration and topic
Source of data	The PD has developed a training program called Sistema.bio Academy which covers the includes essential curriculum with the topic and information every staff member should acquire.
Value(s) applied	40 staff trained
Measurement methods and procedures	Internal records including material (presentations, manuals), and list of participations (digital or paper)
Monitoring frequency	Annual
QA/QC procedures	Human Resources department uses the application BambooHR to store evidences and will ensure veracity and accuracy of the information.
Purpose of data	Ensure all the staff have receive the training required to perform their duties.
Additional comment	

Data / Parameter	Permanent Jobs
Unit	Number of jobs
Description	The project will report the permanent jobs position created.
Source of data	Human Resources records
Value(s) applied	20
Measurement methods and procedures	Internal records including contracts.
Monitoring frequency	Annual

QA/QC procedures	Human Resources department uses the application BambooHR to store evidences and will ensure veracity and accuracy of the information.
Purpose of data	Assessing sustainability
Additional comment	The monitoring frequency will be annual but in case an employee hired works for less than a year, it will be reported but including a note.

In line with the Gold Standard 'Requirements and guidelines: usage rate monitoring v2.0', below, there are further details about "use" and "non-use" definitions applied to define the parameter 'Usage Rates ($U_{p,y}$)':

Step 1. Defining stove use and non-use:	
2.2.3 The project developer shall define project technology "use" versus "non-use" to determine who should be considered eligible for crediting. The definition and criteria applied to define "use" and "non-use" shall be documented in the Project Design Document (PDD) and may be amended with justifications during a given crediting period. Any revisions made shall be documented in the monitoring report along with the justifications.	Biodigester Use: Biodigester works well and is in use; all key aspects in place according to user manual 1. Animals, waste, water available 2. User attests to using the system without reporting lapses in use or technical issues for more than continuous 30 days across the year 3. Operational days counted Non-use: 1. Works well, not in use for more than 30 continuous days 2. Does not work for more than 30 continuous days • Lack of interest • Lack of perceived benefits • There is a problem with the digester • There are no animals • Lack of water • Client moved Biogas Stove Use: Beneficiary confirms they are using biogas stove; hours of biogas stove use per day It is possible that besides the biogas stove another stove (e.g. traditional fogon) is used complementary. Non-use: • The beneficiary states they have stopped using the stove. • The stove is not connected to the biogas hose. • The beneficiary has moved out of the house. • Traditional cookstove other than the biogas stove is in primary use.
2.2.4 To define the use and non-use of project technology, the project developer should use the criteria such as time since last used, frequency of use, duration of use of cooking device, extent to which the traditional technology is displaced etc. The project developer should refer to baseline survey, project survey and Kitchen Performance Tests (KPTs) results to determine the representative cooking practices in the project boundary and identify the criteria for defining use and non-use of project technology.	The project survey includes the following parameters: • Duration of use biogas stove (hours/day) • For other cooking technologies: • Which is the Primary OTHER fuel source? • Which activities do you use the OTHER Energy on? (1) Household activities (2) Productive farm activities • Which is the Secondary OTHER fuel source? • How much wood do you use per month? • How much do you pay for wood per month? • How Much time do you spend collecting firewood each Month • How much LP gas do you use per month • How much do you spend on the LP gas per month

B.7.2. Sampling plan

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The sample size for the annual survey will be estimated based on the simple random sampling outlined in the 'Guideline on Sampling and surveys for CDM project activities and programmes of activities Version 4.0' and methodology 'Technologies and Practices to Displace Decentralized Thermal Energy Consumption' (Version 3.1)

Target population

The monitoring procedure is targeted to be applied on the households, local communities and SMEs installed Sistema.bio's devices, as identified through the Project Database managed by Sistema.bio.

Sampling method

A simple random sampling based on guidelines on "Sampling and surveys for CDM project activities and programme of activities Version 04" will be adopted for estimating the sample size for the monitoring surveys. Simple random sampling is suitable for homogenous populations. Minimum sampling indicated in the 'Technologies and Practices to Displace Decentralized Thermal Energy Consumption' (Version 3.1) will be ensured.

Sample Size:

The minimum total sample size is 100, with at least 30 samples for project technologies of each age being credited. The majority of interviews in a usage survey must be conducted in person and include expert observation by the interviewer within the kitchen in question, while the remainder may be conducted via telephone by the same interviewers on condition that in-kitchen observational interviews are first concluded and analyzed such that typical circumstances are well understood by the telephone interviewers.

Sampling frame: All the households with biogas digester within the project will be the sampling frame.

Data to be monitored:

The necessary data as stated in section B.7.1 above will be collected and monitored by the project proponent as required.

Quality Assurance/Quality Control:

The PO manages its data through a digital application which consists of questions relevant to the monitoring plan as defined in section B.7.1 above. The survey will be performed by the local representative of the PO through a digital app. In order to avoid the possibility of low response rate and answer bias 10% oversampling will be applied by the PO.

Analysis:

The survey data will then be analyzed by the project developer to derive the working status of each biodigester and the consumption of firewood/LPG at the project site (if any). The analysis will form the basis of the monitoring report to be prepared by the developer.

Implementation:

Sistema.bio conducts preparation and pre-testing of the survey questionnaires, in conjunction with the Data and Infrastructure, the Commercial and the Technical Operations teams. Field personnel are trained during their onboarding and then regularly each year through the Sistema.bio Academy that covers all employee training required for each type of role. In addition, the surveys include enumerator prompts and instructions that clearly indicate how questions must be made. These trainings and measures are taken to ensure the highest quality of data possible. The schedule for implementing the sampling effort shall be defined prior to the field activity.

B.7.3. Other elements of monitoring plan

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The project design includes operational and management structure for monitoring, provisions for data archiving, database management (unique identification), and responsibilities and arrangements for data collection and archiving.

The project involved different levels of staff in order to ensure the monitoring activities are performed adequately. The diagram below, summarizes how the responsibilities and task are allocated across the organization structure.

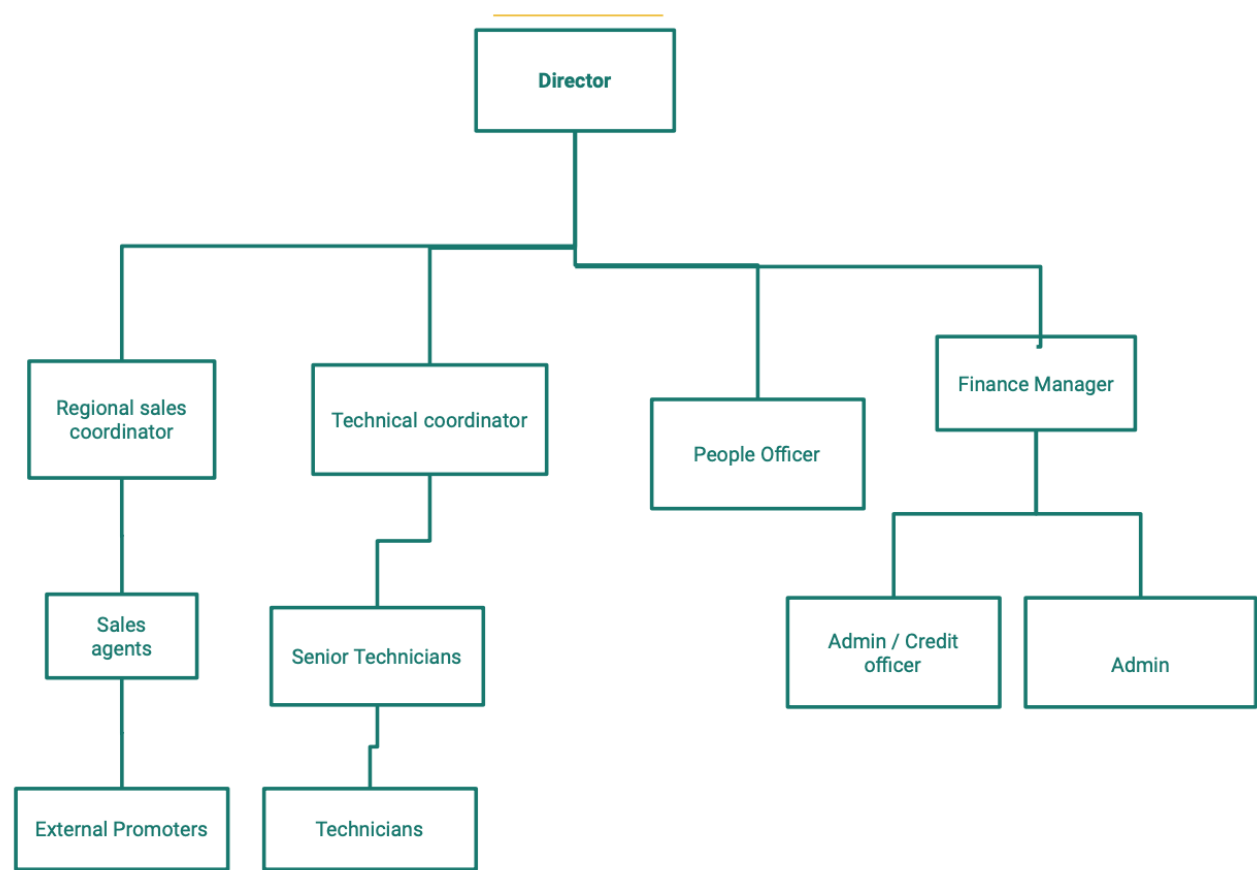


Figure 8 - Organization Chart

Following the methodology requirements, below there are details of the leakage assessment:

Potential source of leakage	Assessment
a) The displaced baseline technologies are reused outside the project boundary in place of lower emitting technology or in a manner suggesting more usage than would have occurred in the absence of the project.	The baseline stoves are not used outside the project boundary, but in some cases, the stove continue being used by the project beneficiaries. The project will account for leakage due to the continued presence of a baseline stove. Although the project’s beneficiaries are encouraged to destroy the baseline stove, some beneficiaries may refuse to destroy the stove. Even destroyed, the construction of an open fire is extremely easy, (e.g it only requires three cinder block or bricks.), therefore, the presence

of a baseline stove will be monitored via the annual monitoring surveys.

This leakage source will be calculated as follows:

Leakage due baselines stove = % of homes that have a stove * net project beneficiaries * cooking time the stove is in use in those HHs * annualized baselines emissions/stove

For biodigesters, there is not risk of leakage identified since the animals remain in the farm that at the project scenario operates with the biodigester technology.

b) Non-project users who previously used lower emitting energy sources use the non-renewable biomass or fossil fuels saved under the project activity.

For the biogas stove, there is no such distinction between a low emitting energy and non-renewable biomass from the firewood consumed in project area. Areas of fuelwood collection, fuelwood suppliers and fuel type are the same for both, project users and non-project users. This potential source of leakage is not considered relevant for the project. For the biodigesters all the baseline scenarios identified means higher emissions.

c) The project significantly impacts the NRB fraction within an area where other CDM or VER project activities account for NRB fraction in their baseline scenario.

The project does not expect to create a negative impact on the NRB; if any, the impact would be positive since the project saves fuelwood by reducing the demand. This potential source of leakage is not considered relevant for the project.

d) The project population compensates for loss of the

The project will carry out an annual survey to assess the leakage due to the replacement of

space heating effect of inefficient technology by adopting some other form of heating or by retaining some use of inefficient technology.

efficient household heating. The project users will be asked if they use the project stove to heat the home outside of regular cooking activity.

This leakage source will be calculated as follows:

Leakage due space heating = % of homes that use open fire stove for heating * net project beneficiaries in operation * time the stove is in use in those HHs for heating * annualized average of baselines emissions/stove.

e) By virtue of promotion and marketing of a new technology with high efficiency, the project stimulates substitution within households who commonly used a technology with relatively lower emissions, in cases where such a trend is not eligible as an evolving baseline.

The fuels and technologies identified at the project scenario are the use of firewood with a traditional stove and/or LPG. The project technology High efficiency technology or renewable energy sources are not commonly used in the project boundary. This potential source of leakage is not considered relevant for the project.

Other potential sources of leakage.

Leakage due to Transportation.

Transportation and maintenance records will be kept. Records include all vehicle types in use by the project at all levels (large trucks, light trucks, cars and motorcycles). Mileage records track miles driven on an ongoing basis for each vehicle, and the results will be tabulated annually. A standard online carbon calculator is used to calculate the total CO₂ produced from driving the total distance driven. That figure is compared against the total emissions being

claimed during the verification period in order to determine leakage.

Leakage due transportation has been calculated using mileage records. The calculations of these leakage source from records available resulted in total emission per years due transportation of 39.62 tCO₂e

SECTION C. DURATION AND CREDITING PERIOD

C.1. Duration of project

C.1.1. Start date of project

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April 1st 2021 is considered as start date of the project. The date represents first batch of biogas digesters installed within the project activity. First installation contracts have been submitted as evidence of the starting date of the project activity. The PD has submitted initial documents for preliminary review on 05/21/2021. Therefore, as per clause 3.4.7 under principle and requirement one year prior to first submission date is taken as start date of the project activity.

C.1.2. Expected operational lifetime of project

>>

15 years

As explained in section A.3 of the PDD, the products are designed to have a long useful lifetime in the harsh, outdoor environments of smallholder farms throughout the world, with a focus on tropical or semi-tropical climates. The reactors are made from high-density polyethylene blends that will retain mechanical and chemical properties in fully exposed conditions for an estimated 20-40+ years. This data has been validated through field data, laboratory exposure testing, certified material characteristics, and supplier claims. The rest of the Sistema.bio components of the kit can be replaced or repaired locally at low costs. Materials that may not last 20 years, like hose clamps, PVC piping, or gas tubing can be easily replaced on the unit from local materials with local know-how and with technical support as needed. In addition, most of the components, including the reactor, can be repaired or replaced in the field, ensuring continued operations over time.

The supplier of the geomembrane gives a warranty of 10 years and the ISO certification of the assembly procedure ensures the quality and lifetime of the reactor.

The project beneficiary is responsible to replace components of the kit (hose clamps and, PVC piping, and filling material of biogas filter). The project is considering supply those components when the beneficiary cannot afford the spare parts to ensure continuous operation of the biodigesters.

The biogas stove is made of long-lasting materials (carbon steel, electrostatic painting, aluminum). The stove has a 3-years warranty, but the expected lifetime is longer. The stove's emissions and performance have been tested in accordance with ISO 19867-1. Older stoves tested are less than 3 years old; no field test is available to corroborate the exact lifetime, only empirical conclusion regarding the lifespan of the stove when maintenance is done as stipulated in the manual. The project will evaluate the conditions of the stove in the 3rd year to define degradation and will replace the components as needed to ensure continuous operation. In any case, Sistema.bio will ensure the operational lifetime of the stove. The statement below provides further information about the continuous operation of the project technology.

The conditions signed and agreed by the technology users stipulate the continuous use of the technology, conduct timely maintenance and replace components of the kit as needed (hose clamps, PVC piping, or gas tubing can be easily replaced on the unit from local materials with local know-how and with technical support as needed).

Reactors and stoves can be relocated, refurbished, and redeployed. This ensures that if there are siting conflicts, or if a user desires to move, the equipment can continue to work against its service life. All the units relocated to a different user will be identified transparently in the sales records. A check to prevent double-counting will be done to ensure the same serial number of equipment is not assigned simultaneously to more than one user (e.g. marked as dropped-off under the first user record, and registered under the new user record).

C.2. Crediting period of project

C.2.1. Start date of crediting period

>>

April 1st 2022

C.2.2. Total length of crediting period

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5 years renewable

SECTION D. SUMMARY OF SAFEGUARDING PRINCIPLES AND GENDER SENSITIVE ASSESSMENT

D.1. Safeguarding Principles that will be monitored

A completed Safeguarding Principles Assessment is in [Appendix 1](#), ongoing monitoring is summarised below.

Principles	Mitigation Measures added to the Monitoring Plan
N/A	No mitigation measures were identified during the stakeholder consultation.

D.2. Assessment that project complies with GS4GG Gender Sensitive requirements

Question 1 - Explain how the project reflects the key issues and requirements of Gender Sensitive design and implementation as outlined in the Gender Policy?	The main entity in Mexico that watches over human rights is the National Commission of Human Rights³⁷ (CNDH for its meaning in Spanish). The CNDH design policies and indicators to ensure the government entities respect human rights. Also, assists victims of human right offenses. The scope of action of CNDH is limited to the government (in all levels Federal, State and Municipality) performance. The main action from CNDH regarding gender equality is the 'Contribution Compliance Index of National Policy between Women and Men'.³⁸ This index includes: 1) public policies, 2) legislative monitoring, 3) politic participation, and 4) human rights promotion. As explained, the dimension of this index only includes government performance; the index does not relate private sector. The Analysis of Compliance State of Human Rights from Business Activities³⁹ documented 65 recommendations issued between 1990 and 2018 due human rights violations in the business context. As can be seen, the incidence of documented cases of human rights violations from business activities is
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³⁷ <http://informe.cndh.org.mx/Default.aspx>

³⁸ http://informe.cndh.org.mx/images/uploads/nodos/61084/content/files/Nota_indicador.pdf

³⁹ <http://informe.cndh.org.mx/menu.aspx?id=60113>

	minimum, it may explain the absence of specific recommendation and policies for private sector. The project respects the key gender issues and requirements of gender-sensitive design and implementation of the project. The project is aimed to avail the clean cooking solutions to the households. Biogas project will result in displacing firewood consumption. Therefore, the project will reduce consumption of firewood. In Mexico, the majority of chores around the household, and especially in the kitchen, are handled by women. This exposes them to indoor air pollution when using fuelwood, and the associated health issues that come with that.
Question 2 - Explain how the project aligns with existing country policies, strategies and best practices	As explained in Question 1, there is no specific requirements or policies mandatory for the business sectors issued by the CNDH. Instead, the project has identified the recommendation from the United Nations (ONU Mujeres) relevant for the region. Those recommendation aim to end violence against women in Latin America and Caribbean.⁴⁰ Specifically, the project adheres to the general recommendations 12 and 19 adopted by the committee on the elimination of discrimination against women.⁴¹
Question 3 - Is an Expert required for the Gender Safeguarding Principles & Requirements?	No, an expert is not required for Gender Safeguarding Principles and Requirements. We address Gender requirements in the Safeguarding principles assessment.

⁴⁰ <https://lac.unwomen.org/es/que-hacemos/fin-a-la-violencia-contra-las-mujeres>

⁴¹ https://tbinternet.ohchr.org/Treaties/CEDAW/Shared%20Documents/1_Global/INT_CEDAW_GEC_5831_E.pdf; https://tbinternet.ohchr.org/Treaties/CEDAW/Shared%20Documents/1_Global/INT_CEDAW_GEC_3731_E.pdf

Question 4 - Is an Expert required to assist with Gender issues at the Stakeholder Consultation?	No, an expert will not be required to assist with gender issues at the Stakeholder Consultation
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SECTION E. SUMMARY OF LOCAL STAKEHOLDER CONSULTATION

The below is a summary of the 2 step GS4GG Consultation for monitoring purposes. Please refer to the separate Stakeholder Consultation Report for a complete report on the initial consultation and stakeholder feedback round.

E.1. Summary of stakeholder mitigation measures

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Although the stakeholder consultation included an assessment of the safeguarding principles, no mitigation measures were identified during the stakeholder consultations. The LSC report includes details about comments received during the different meeting organized (online, in-person and SFR).

E.2. Final continuous input / grievance mechanism

Method	Include all details of Chosen Method (s) so that they may be understood and, where relevant, used by readers.
Continuous Input / Grievance Expression Process Book (mandatory)	Location: Puebla 124, Roma Norte, Ciudad de México, México. The input and grievance expression can be reflected via telephone, email or writing on grievance book. All the contact information was announced to stakeholders
GS Contact (mandatory)	help@goldstandard.org
Telephone access (optional)	+52 55 52 11 86 17
Internet/email access (optional)	partnership@sistema.bio

APPENDIX 1 - SAFEGUARDING PRINCIPLES ASSESSMENT

Complete the Assessment below and copy all Mitigation Measures for each Principle into [SECTION D](#) above. Please refer to the instructions in the [Guide to Completing](#) this Form below.

Assessment Questions/ Requirements	Justification of Relevance (Yes/potentially/no)	How Project will achieve Requirements through design, management or risk mitigation.	Mitigation Measures added to the Monitoring Plan (if required)
Principle 1. Human Rights			
1. The Project Developer and the Project shall respect internationally proclaimed human rights and shall not be complicit in violence or human rights abuses of any kind as defined in the Universal Declaration of Human Rights 2. The Project shall not discriminate with regards to participation and inclusion	1. No 2. No	1. The project is implemented respecting internationally proclaimed human rights and is no complicit in violence or human rights abuses of any kind as defined in the Universal Declaration of Human Rights 2. The project doesn't either discriminate with regards to participation and inclusion as the biodigesters are distributed in collaboration with the representatives of the local communities. Sistema.bio has in place organizational policies to	N/A

		address respect to human rights. Specifically, there is an internal 'Social policy and procedures'.⁴² This policy describes principles and targets relating the relationship with employees and society including (among other): Conclusion: the parameter will not be monitored.	
Principle 2. Gender Equality			
1. The Project shall not directly or indirectly lead to/contribute to adverse impacts on gender equality and/or the situation of women 2. Projects shall apply the principles of nondiscrimination, equal treatment, and equal pay for equal work 3. The Project shall refer to the country's national		1. The project activity doesn't endorse any form of discrimination based on gender. Every beneficiary decides if they want the project biodigester /biogas stove. It's not foreseen that the project reduces or put at risk women's access to or control of resources, entitlements and benefits. Instead, it's foreseen that the women, as main	N/A

⁴² Política Social_Sistema.bio.pdf

gender strategy or equivalent national commitment to aid in assessing gender risks 4. (where required) Summary of opinions and recommendations of an Expert Stakeholder(s)		responsible for firewood collection and cooking activities will have better control of resources (firewood and time will be saved) and moreover to benefit most for the possible health improvements caused by the reduced smoke inhalation during the cooking activities. It is thereafter not either foreseen that the Project would adversely affect men and women in marginalized or vulnerable communities. There will be less burden on women, men and children, as less firewood for cooking needs to be collected. This will reduce the time burden on women and men in the socially isolating activity of collecting resources. The Project takes into account gender roles and the abilities of women and men to participate in the	
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		decision/designs of the project activities. For example, the Stakeholder Consultation made included both women and men participating in the consultation meeting. In fact, women's participation and engagement in the project (as they are the main responsible of the cooking activities) will be essential for the success of the whole project. The Project will take into account gender roles and the abilities of women and men to participate and benefit from the project activities. For example, the training/cooking demonstrations on using the new stoves and on their benefits will be targeted especially for women who are traditionally responsible for the cooking activities.	
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		The project is not contributing to an increase in women's workload or preventing them from engaging in other activities. In fact, the biodigester will reduce the firewood need for daily cooking activities and will thereafter reduce women's and girls' workload related to the firewood collection. The project is not foreseen to reproduce or deepen discrimination against women. The women's role will be essential as the biodigester users and they will have the possibility for giving feedback regarding the project as any other community member. The project is not foreseen to limit women's ability to use, develop and protect natural resources. Instead, the use of biodigesters will reduce the firewood	
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		consumption and will thereafter give a possibility for saving and local natural wood resources. The project activity will not expose women or girls to further risks or hazards. Instead, the risk related to the smoke inhalation during the cooking activities or the risks related to the firewood collection are foreseen to be reduced. 2. The Project will not have any impact on women's ownership rights to inherit and own land, homes and other assets. The Project applies the principles of non-discrimination and equal treatment and equal pay for equal work. Sistema.bio has in place organizational policies to address respect gender equality. Specifically, there	
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		is in place an internal gender policy.⁴³ The Sistema.Bio Gender policy includes the following principles: • Equal pay for men and women • Non-discrimination in recruitment and non-termination based on gender • Non-discrimination in job assignment or promotion based on gender • Strict zero tolerance towards sexual harassment in any instance in all of our offices and among all of our employees • Trainings on gender issues led by men and women every 6 months • Incentive for feedback from both genders	
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⁴³ Política de Género_Sistema.bio.pdf

		regarding their experience working with Sistema.bio • Adjusting policies, benefits, and work environments to empower women, helping them thrive with us and reducing gender-specific barriers such as: ○ With generous leave policies - from unlimited vacation to generous maternity and paternity leave. ○ With remote work flexibility and work schedules when needed so that employees can take care of themselves and their families. ○ Respecting gender requirements with breastfeeding spaces dedicated to women with children. For the project monitoring activities and for any other eventual paid or volunteer	
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		work the principle of the equal pay for equal work will be applied and it will be organized in way to provide the conditions for equitable participation of men and women whenever possible. Project activities like using the biogas cookstoves is not having any limitations on participation or benefiting from the Project depending on the pregnancy, maternity/paternity leave or marital status. 3. The Project will not include any specific gender related risks. The project follows the basic gender assessment required by the standard. 4. Not applicable. GS didn't require at Preliminary Review that Project seeks the input of an Expert Stakeholder.	
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		Conclusion: the parameter will not be monitored.	
Principle 3. Community Health, Safety and Working Conditions			
1. The Project shall avoid community exposure to increased health risks and shall not adversely affect the health of the workers and the community	1. No	The project activities don't include exposing the community to increased health risks and is not adversely affecting the health of workers and the community. The biogas generation and storage at the biodigester does not represent a risk for explosion. There are no precedents of these kind of incidents in biodigesters. The biogas density tends to be lower than air, therefore, in case of leakage, it is dissipated easily and its combination with air (oxygen) is never enough to make it explosive.	N/A

		The project includes a training and extensive explanation about safety and care for users. The directions on how to use the biodigester and the biogas are delivered in written manner ⁴⁴, including directions how to handle and use the biof-fertilizer (biol)⁴⁵, and safety measure to prevent accidents (e.g. risk of exposure to H2S). Every usure should confirm attendance of traing.⁴⁶ The PD has developed a training program called Sistema.bio has developed an internal project called Sistema.bio Academy that covers all employee training required for each type of role. Once an employee starts working at Sistema.bio, they receive a curriculum designed	
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⁴⁴ 1901_MEXICO_MANUAL_DE_USUARIO.pdf

⁴⁵ 2021_ Sistema.bio_Manual Biofertilizante usuarios.pdf

⁴⁶ b.HS_M1_ENG_R&P Commitment letter.pdf

		for them to learn everything they need to know about their job. Employees who will be working directly with farmers and systems in the field take a very comprehensive Health and Safety module and a Risk Prevention model, and they are required to sign an agreement stating they have received the training. In addition, Sistema.bio has enacted more general policies for all staff in relation to safe working conditions.⁴⁷ Beside the extensive training to project staff, Sistema.bio has in place policies to ensure health and safety for all the project employees.⁴⁸ Every training session for project staff is documented.⁴⁹ Sistema.bio has in place a code of conduct the explain	
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⁴⁷ Sistema.bio Academy Summary.pdf

⁴⁸ Samples of training presentation of H&S: 'a.HS_M1_ENG_Risk & prevention AD.pdf', 'a.HS_M2_ENG_Health & Safety in the field.pdf'.

⁴⁹ Template of Agreement: Reception of the following trainings : HEALTH & SAFETY and RISK OF ANAEROBIC DIGESTION (see file: b.HS_M2_ENG_H&S Agreement.pdf)

		the professional practices to followed by all the project staff.⁵⁰ Recently, Sistema.bio incorporated a 'Psychosocial Risk Prevention Policy' aligned with the Mexican Norm Number 035 (NOM035 in Spanish).⁵¹ The aim of this norm is for the employer to identify and analyze psychosocial and occupational risk factors that are directly related to the health of employees. Sistema.bio has in place other policies⁵² in place including: - Environmental - Anticorruption - Social In fact, the project activity provides the installation of	
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⁵⁰ Manual de empleadas y empleados_Sistema.bio.pdf

⁵¹ NOM.035 Prevención de Riesgos Psicosociales_Sistema.bio.pdf

⁵² Política Ambiental Sistema.bio.pdf, Política Antisoborno y Anticorrupción_Sistema.bio.pdf, Política Social_Sistema.bio.pdf

		biodigesters and distribution of biogas stoves, with the aim, omitting the other objectives, to improve the health of households, for example through the reduction of smoke and therefore less harmful inhalations, odor and other contamination vectors. Conclusion: the parameter will not be monitored.	
Principle 4.1 Sites of Cultural and Historical Heritage			
Does the Project Area include sites, structures, or objects with historical, cultural, artistic, traditional or religious values or intangible forms of culture?	No	The project activity doesn't include sites, structures or objects with historical, cultural, artistic, traditional or religious value or intangible forms of culture. The Project will provide biodigesters and biogas stoves to the households in the project area and it does not require alteration, damage or removal of any historical, artistic, traditional, religious or cultural heritage issues.	N/A
>>			

		Conclusion: the parameter will not be monitored.	
Principle 4.2 Forced Eviction and Displacement			
Does the Project require or cause the physical or economic relocation of peoples (temporary or permanent, full or partial)?	No	The project activity consists of distributing biodigesters and therefore no physical or economic relocation of peoples is involved. Conclusion: the parameter will not be monitored.	N/A
>>			
Principle 4.2 Forced Eviction and Displacement			
Does the Project require or cause the physical or economic relocation of peoples (temporary or permanent, full or partial)?	a. No b. No	Biodigester distribution doesn't need additional lands to be used and, therefore, doesn't require any change to land tenure arrangements and/or other rights. In fact, the aim of the project is to reduce the quantity of firewood consumed in the project area. Conclusion: the parameter will not be monitored.	N/A
>>			
Principle 4.3 Land Tenure and Other Rights			
Does the Project require any change, or have any	No	The project does not represent a harm for	N/A

uncertainties related to land tenure arrangements and/or access rights, usage rights or land ownership?		indigenous people present within the area of influence. The project does not make a difference based on ethnic or race origin. Conclusion: the parameter will not be monitored.	
>>			
Principle 4.4 - Indigenous people			
Are indigenous peoples present in or within the area of influence of the Project and/or is the Project located on land/territory claimed by indigenous peoples?	No	The project does not represent a harm for indigenous people present within the area of influence. The project does not make a difference based on ethnic or race origin. Conclusion: the parameter will not be monitored.	N/A
Principle 5. Corruption			
1. The Project shall not involve, be complicit in or inadvertently contribute to or reinforce corruption or corrupt Projects	No	The Project doesn't involve, complicit in or inadvertently contribute to or reinforce corruption or corrupt projects. Conclusion: the parameter will not be monitored.	N/A
Principle 6.1 Labour Rights			

1. The Project Developer shall ensure that all employment is in compliance with national labour occupational health and safety laws and with the principles and standards embodied in the ILO fundamental conventions 2. Workers shall be able to establish and join labour organisations 3. Working agreements with all individual workers shall be documented and implemented and include: a) Working hours (must not exceed 48 hours per week on a regular basis), AND b) Duties and tasks, AND c) Remuneration (must include provision for	1. No 2. No 3. No 4. No 5. No	1. The project is implemented in the field by Sistema.bio. The employees' rights are a cross-cutting issue and respected in all parts of the projects and project partners. Sistema.bio employees, project subcontractors, and all project staff will work voluntarily for the project, no forced labour is used and all employment is in compliance with national laws and consistency with the principles of standard ILO conventions. Samples of the fulltime and part time employees' contracts have been made available to the VVB to ensure alignment with requirement from host country.⁵³ Sistema.bio has also enacted a Gender Policy that outlines	Training offered to project staff will be monitored and reported.
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⁵³ CONTRATO INDIVIDUAL DE TRABAJO POR TIEMPO INDEFINIDO - TECHOPS-SEPT2021.docx, CONTRATO INDIVIDUAL DE TRABAJO POR TIEMPO INDEFINIDO_SEPT2021.docx, CONTRATO INDIVIDUAL TRABAJO -VENTAS-SEPT2021.docx

payment of overtime), AND d) Modalities on health insurance, AND e) Modalities on termination of the contract with provision for voluntary resignation by employee, AND f) Provision for annual leave of not less than 10 days per year, not including sick and casual leave. 4. No child labour is allowed (Exceptions for children working on their families' property requires an Expert Stakeholder opinion) 5. The Project Developer shall ensure the use of appropriate equipment, training of workers, documentation and reporting of accidents and incidents, and		the vision regarding gender equality and the specific actions taken in the company to ensure equal representation for all employees. Finally, Sistema.bio complies with all labor norms in the country and goes beyond the minimum expected. The information relating Health, Safety and Working Conditions and project alignment with local requirements are summarized above in Safeguarding principle 3. 2. The workers can establish and join labour organizations. In case new workers will be hired, the working agreement will be documented and implemented in compliance with the Section 3.6.1 of	
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emergency preparedness and response measures		GS4GG Safeguarding Principles & Requirements version 1.1. The employment model applied will be also locally and culturally appropriate. 3. The Project staff are aligned with local legislation and meet all the requirements relating working hours, remuneration, health insurance and leaves. The project staff receives all the benefits required by the law. The contracts aligned with local legislation states the duties and task; modalities of termination including provision for voluntary resignation; and provision for annual leaves. From the stove and biodigester users perspective: The use of the biogas cookstove will displace the firewood used in daily cooking activities. The use of	
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		biodigesters will reduce the time and effort to treat the manure and can thereafter release economical and time resources of the local families for other tasks which can be considered to support the financial sustainability of the project. 4. The Project only employs adults with legal age to work. The project does involve any child labor. 5. The project provides all the training, equipment and procedure to prevent accidents and plan in case any incident. Conclusion: the parameter will not be monitored.	
Principle 6.2 Negative Economic Consequences			
1. Does the project cause negative economic consequences during and after project implementation?	1. No	The financial model of the project is based on the carbon credits revenues and is channeling the resources to ensure the project	N/A

>>		implementation and monitoring. The project serves vulnerable and marginalized communities. No potential risks for the local economy are expected. The project will not cause negative economic consequences during implementation nor during its operation. Conclusion: the parameter will not be monitored.	
Principle 7.1 Emissions			
Will the Project increase greenhouse gas emissions over the Baseline Scenario?	No	The Project will reduce the GHG emissions as will be monitored and verified in line with the GS4GG. Conclusion: the parameter will not be monitored.	N/A
>>			
Principle 7.2 Energy Supply			
Will the Project use energy from a local grid or power supply (i.e., not connected to	No	The Project will not use energy from a local grid or power supply. The	N/A

a national or regional grid) or fuel resource (such as wood, biomass) that provides for other local users?		biodigesters produce biogas and therefore there will be a change in the cooking fuel, this biogas will be produced from the waste in each household. Conclusion: the parameter will not be monitored.	
>>			
Principle 8.1 Impact on Natural Water Patterns/Flows			
Will the Project affect the natural or pre-existing pattern of watercourses, ground-water and/or the watershed(s) such as high seasonal flow variability, flooding potential, lack of aquatic connectivity or water scarcity?	No	The project will not affect the natural or pre-existing pattern of watercourses, groundwater and/or the watersheds etc. water related issues. Conclusion: the parameter will not be monitored.	N/A
>>			
Principle 8.2 Erosion and/or Water Body Instability			
Could the Project directly or indirectly cause additional erosion and/or water body instability or disrupt the natural pattern of erosion?	a. No b. No	The Project will not cause additional erosion directly or indirectly and/or water body instability or disrupt the natural pattern of erosion. Conclusion: the parameter will not be monitored.	N/A
>>			

Principle 9.1 Landscape Modification and Soil			
Does the Project involve the use of land and soil for production of crops or other products?	No	The project impact on the environment is positive, no negative impacts are expected. Moreover, the stove distribution activities do not include planting or other agricultural activities, producing chemicals or use of GMOs. The project will distribute biodigesters produced locally. The local biodigester production does not have any significant environmental impacts. Neither hazardous waste is produced. Conclusion: the parameter will not be monitored.	N/A
>>			
Principle 9.2 Vulnerability to Natural Disaster			
Will the Project be susceptible to or lead to increased vulnerability to wind, earthquakes, subsidence, landslides, erosion, flooding, drought or other extreme climatic conditions?	No	The Project is not suspected to or will lead to increased vulnerability to any extreme climatic conditions. Furthermore, the aim of the project is to reduce the quantity of firewood and LPG	N/A

>>		consumed in the project area for cooking activities which will save the natural resources. Conclusion: the parameter will not be monitored.	
Principle 9.3 Genetic Resources			
Could the Project be negatively impacted by or involve genetically modified organisms or GMOs (e.g., contamination, collection and/or harvesting, commercial development, or take place in facilities or farms that include GMOs in their processes and production)?	No	The Project doesn't involve or is negatively impacted by the use of genetically modified organisms or GMOs. Conclusion: the parameter will not be monitored.	N/A
>>			
Principle 9.4 Release of pollutants			
Could the Project potentially result in the release of pollutants to the environment?	No	The Project is not potentially resulting in release of pollutants to the environment. Conclusion: the parameter will not be monitored.	N/A
>>			
Principle 9.5 Hazardous and Non-hazardous Waste			
Will the Project involve the manufacture, trade, release,	No	The Project is not involving the manufacture, trade,	N/A

and/ or use of hazardous and non-hazardous chemicals and/or materials?		release, and/or use of hazardous chemicals and or materials. Conclusion: the parameter will not be monitored.	
>>			
Principle 9.6 Pesticides & Fertilisers			
Will the Project involve the application of pesticides and/or fertilisers?	No	The Project doesn't involve the application of pesticides and/or fertilizers. Conclusion: the parameter will not be monitored.	N/A
>>			
Principle 9.7 Harvesting of Forests			
Will the Project involve the harvesting of forests?	No	The Project doesn't involve harvesting of forests. The aim of the project is to reduce the quantity of firewood consumed, therefor reducing pressure of biomass harvest. Conclusion: the parameter will not be monitored.	
>>			
Principle 9.8 Food			
Does the Project modify the quantity or nutritional quality of food available such as through crop regime alteration	No	The Project doesn't modify the quantity or nutritional quality of food available.	N/A

or export or economic incentives?		The stove models include two burners, which represent an opportunity to cook simultaneously more than one dish expanding cooking options.	
>>		Conclusion: the parameter will not be monitored.	
Principle 9.9 Animal husbandry			
Will the Project involve animal husbandry?	No	The Project doesn't involve animal husbandry.	N/A
>>		Conclusion: the parameter will not be monitored.	
Principle 9.10 High Conservation Value Areas and Critical Habitats			
Does the Project physically affect or alter largely intact or High Conservation Value (HCV) ecosystems, critical habitats, landscapes, key biodiversity areas or sites identified?	No	The project boundary is the physical, geographical sites of the project technologies, in other words, the physical location of the project stoves. The stoves are installed only at the households and there are no interventions in HCV areas. Therefore, there is no need to carry out a HCV assessment.	N/A
>>		Conclusion: the parameter will not be monitored.	

Principle 9.11 Endangered Species			
Are there any endangered species identified as potentially being present within the Project boundary (including those that may route through the area)? AND/OR Does the Project potentially impact other areas where endangered species may be present through transboundary affects?	No	The project boundary is the physical, geographical sites of the project technologies, in other words, the physical location of the biodigesters. There are no endangered species identified as potentially being present in the project boundary and the project is not foreseen to have any negative potential impacts on other areas where endangered species may be present through transboundary effects. Conclusion: the parameter will not be monitored.	N/A
>>			

APPENDIX 2- CONTACT INFORMATION OF PROJECT PARTICIPANTS

Organization name	Buen Manejo del Campo S.A de C.V (Sistema.bio)
Registration number with relevant authority	RFC: BMC100326DL5
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APPENDIX 3- LUF ADDITIONAL INFORMATION

Risk of change to the Project Area during Project Certification Period:	
Risk of change to the Project activities during Project Certification Period:	
Land-use history and current status of Project Area:	
Socio-Economic history:	
Forest management applied (past and future)	
Forest characteristics (including main tree species planted)	
Main social impacts (risks and benefits)	
Main environmental impacts (risks and benefits)	
Financial structure	
Infrastructure (roads/houses etc):	
Water bodies:	
Sites with special significance for indigenous people and local communities - resulting from the Stakeholder Consultation:	
Where indigenous people and local communities are situated:	
Where indigenous people and local communities have legal rights, customary rights or sites with special cultural, ecological, economic, religious or spiritual significance:	

APPENDIX 4-SUMMARY OF APPROVED DESIGN CHANGES

Please refer to Annex A of [Principles and Requirements](#) for more information on procedures governing Design Changes

Revision History

Version	Date	Remarks
1.2	14 October 2020	Hyperlinked section summary to enable quick access to key sections Improved clarity on Key Project Information Inclusion criteria table added Gender sensitive requirements added Prior consideration (1 yr rule) and Ongoing Financial Need added Safeguard Principles Assessment as annex and a new section to include applicable safeguards for clarity Improved Clarity on SDG contribution/SDG Impact term used throughout Clarity on Stakeholder Consultation information required Provision of an accompanying Guide to help the user understand detailed rules and requirements
1.1	24 August 2017	Updated to include section A.8 on 'gender sensitive' requirements
1.0	10 July 2017	Initial adoption